

6

Inventory and Merchandising Operations



SPOTLIGHT | Fast Retailing www.fastretailing.com

Fast Retailing (FR), a public Japanese retail holding company, is the company responsible for the very successful UNIQLO brand, as well as others such as GU, Theory, Comptoir des Cottonniers, Princesse tam.tam, and J Brand. Since the inauguration of its first store in Hiroshima in 1984, UNIQLO's growth in Japan, and later regionally and globally, has been the envy of many retailers. Now Asia's biggest clothing retailer with sales of about US \$15 billion, FR is only behind Inditex (Zara) and H&M. FR's Chief Executive, Tadashi Yanai, has set a lofty goal of making the company the world's top apparel retailer by 2020.

As a retailer, inventory is an important part of FR's operations. Its Balance Sheet showed that inventories (at ¥260 billion, about US \$2.2 billion) comprise the second largest current asset after cash at ¥355 billion. In FR's inventory policy, on the next page, certain terms, such as *purchase allowance* and *consignment*, may be new to you now, but as you go through the chapter, you will understand how FR determines the amount it reports as inventory on its Balance Sheet. ●



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ADAPTED EXCERPTS FROM FAST RETAILING'S NOTES TO THE ACCOUNTS

Inventories

Inventories are valued at the lower of cost and net realizable value; the weighted average method is principally used to determine cost. Net realizable value is based on the estimated selling price in the ordinary course of business less any estimated costs to be incurred to sell the goods.

Source: From Fast Retailing, Annual Report, 2015, page 10.

A1			
	A	B	C
1	Fast Retailing Consolidated Balance Sheet	At August 31	
2	(Adapted, in billions of ¥)	2015	2014
3	Cash at banks	¥ 355.2	¥ 314.0
4	Trade and other receivables	44.8	47.4
5	Inventories	260.0	223.2
6	All other current assets	214.4	132.3
7	Total current assets	874.4	717.0
8	Total non-current assets	289.3	275.3
9	Total assets	1,163.7	992.3
10	Total liabilities	388.9	356.3
11	Total equity	774.8	636.0
12	Total liabilities and equity	¥ 1,163.7	¥ 992.3
13			

In addition to this, cost of sales (¥833 billion) was the biggest expense, on net sales of around ¥1,682 billion, on FR's Consolidated Income Statement for the financial year ended August 31, 2015. How was this cost of sales arrived at?

A1			
	A	B	C
1	Fast Retailing Consolidated Income Statement	12 Months Ended Aug. 31	
2	(Adapted, in billions of ¥)	2015	2014
3	Revenue	¥ 1,681.8	¥ 1,382.9
4	Cost of sales	(833.2)	(683.2)
5	Gross profit	848.6	699.7
6	Selling, general, and administrative expenses	(671.9)	(549.2)
7	Other income	8.8	7.0
8	Other expenses	(21.0)	(27.2)
9	Operating profit	164.5	130.3
10	Finance income	17.4	6.0
11	Finance costs	(1.1)	(0.9)
12	Income taxes	(63.3)	(56.1)
13	Profit for the year	¥ 117.5	¥ 79.3
14			

You can see that cost of sales is by far FR's largest expense item. The title *Cost of Merchandise*, *Cost of Sales*, or *Cost of Goods Sold* perfectly describes that expense. In short:

- Fast Retailing buys (or manufactures) inventory, an asset carried on the books at cost.
- Once sold, the inventory is no longer FR's assets. The cost of inventory sold gets shifted into the expense account, called Cost of Sales or Cost of Goods Sold (usually abbreviated COGS). We will use both terms interchangeably.

Inventory is the heart of a merchandising business, and cost of goods sold is the most important expense item for a retailer. Gross profit (or gross margin, line 5 in FR's Consolidated Income Statement) is the difference between net sales and cost of goods sold. This chapter covers the accounting for inventory and cost of goods sold. It also shows you how to further analyze financial statements. Here we focus on inventory, cost of goods sold, and gross profit.

LEARNING OBJECTIVES

- | | |
|---|---|
| <p>1 Understand the nature of inventory and retailing operations</p> <p>2 Record inventory-related transactions</p> <p>3 Understand and apply different inventory cost assumptions</p> | <p>4 Analyze effects of inventory errors</p> <p>5 Evaluate a company's retailing operations</p> |
|---|---|

UNDERSTAND THE NATURE OF INVENTORY AND RETAILING OPERATIONS

- 1 Understand** the nature of inventory and retailing operations

We begin by showing how the financial statements of merchandisers or retailers, such as Fast Retailing or Muji, differ from those of service entities, such as Qatar Airways, and accounting firms. The financial statements in Exhibit 6-1 highlight how service entities differ from merchandisers (dollar amounts are assumed).

The basic concept of accounting for merchandise inventory can be illustrated with an example. Suppose a retailer has in stock three shirts that cost \$30 each. The retailer sells two shirts for \$50 each:

- The Balance Sheet reports the one shirt that the company still holds in inventory.
- The Income Statement reports the cost of the two shirts sold, as shown in Exhibit 6-2.

This is the basic concept of how we identify **inventory**, the asset, from **cost of goods sold** (COGS) or **cost of sales**, the expense. The cost of the inventory sold shifts from asset to expense when the seller delivers the goods to the buyer. We discussed this in Chapter 3, and looked at the revenue recognition criteria stipulated in *IAS 18—Revenue*. For manufacturers, inventory is handled in much the same way as merchandisers, the difference being that manufacturers buy raw materials (accounted for as “raw materials inventory”) that they work on (as “work-in-progress, or WIP inventory”) until they are ready for sale (“finished goods inventory”). Whatever costs (and overheads) they incur during the production process is added (or allocated) to the WIP inventory. Finished goods that are sold are shown in the Income Statement, and those that remain on hand are shown in the Balance Sheet (Exhibit 6-3).

Exhibit 6-1 Contrasting a Service Company with a Merchandiser

Service Company			Merchandising Company		
Qatar Airways Income Statement			Fast Retailing Income Statement		
1			1		
2	Service revenue	\$ XXX	2	Sales revenue	\$ XXX
3	Expenses:		3	Cost of goods sold	XXX
4	Salary expense	X	4	Gross profit	XXX
5	Depreciation expense	X	5	Operating expenses:	
6	Income tax expense	X	6	Salary expense	X
7	Net income	\$ X	7	Depreciation expense	X
8			8	Income tax expense	X
			9	Net income (net loss)	\$ X
			10		

Qatar Airways Balance Sheet			Fast Retailing Balance Sheet		
1			1		
2	Assets		2	Assets	
3			3		
4	Current assets		4	Current assets	
5	Cash	\$ X	5	Cash	\$ X
6	Accounts receivable, net	X	6	Accounts receivable, net	X
7	Prepaid expenses	X	7	Inventory	X
8			8	Prepaid expenses	X
			9		

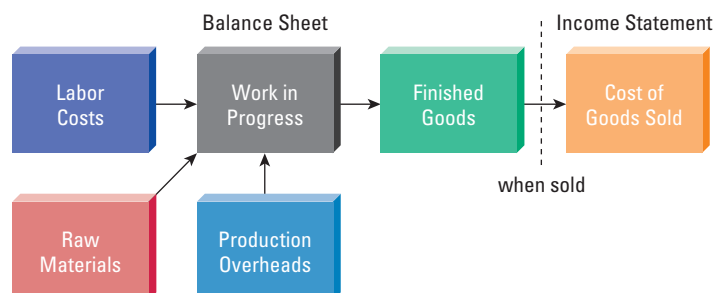
Exhibit 6-2 Inventory and Cost of Goods Sold When Inventory Cost Is Constant

Balance Sheet (partial)

Current assets	
Cash.....	\$XXX
Short-term investments	XXX
Accounts receivable.....	XXX
Inventory (1 shirt @ cost of \$30).....	30
Prepaid expenses	XXX

Income Statement (partial)

Sales revenue	
(2 shirts @ sale price of \$50 each)	\$100
Cost of goods sold	
(2 shirts @ cost of \$30 each)	60
Gross profit	\$ 40

Exhibit 6-3 Overview of Inventory for Manufacturers

A retailer is more likely to have only finished goods in its inventory, whereas a manufacturer may have inventories at various stages of completion (from raw materials to finished goods). We shall limit our discussion in this chapter to finished goods, or goods ready for sale, but the principles of accounting for inventory remain the same for all three categories of inventories. Remember how we earlier spoke of management accounting? The topic of product costing and cost allocation methods are major components of a typical management accounting course.

Sale Price vs. Cost of Inventory

Note the difference between the sale price of inventory and the cost of inventory. In our earlier example:

- Sales revenue is based on the *sale price* of inventory sold (\$50 per shirt).
- Cost of goods sold is based on the *cost* of inventory sold (\$30 per shirt).
- Inventory on the Balance Sheet is based on the *cost* of inventory still on hand (\$30 per shirt).

Gross profit, also called **gross margin**, is the excess of sales revenue over cost of goods sold. It is called *gross* profit because operating expenses have not yet been subtracted. Fast Retailing’s gross margin was ¥848.6 billion (i.e., net sales of ¥1,681.8 billion less cost of sales of ¥833.2 billion) as shown earlier. On its Balance Sheet, remaining inventory at the end of the financial year totaled ¥260 billion.

This ending inventory represents:

Inventory
(Balance Sheet)

=

Number of units of
inventory *on hand*

×

Cost per unit
of inventory

And the cost of goods sold of ¥833.2 billion represents:

Cost of goods sold
(Income Statement)

=

Number of units of
inventory *sold*

×

Cost per unit
of inventory

Let’s see what “units of inventory” and “cost per unit” mean.

Number of Units of Inventory. The number of inventory units on hand is determined from the accounting records, backed up by a physical count of the goods at year-end. Companies do not include any goods in their inventory that they hold on **consignment** because those goods belong to another company. But they do include their own inventory that is out on consignment and held by another company. Companies include inventory in transit from suppliers or in transit to customers that, according to shipping terms, legally belong to them as of the year-end. Shipping terms, otherwise known as *FOB terms*, indicate who owns the goods at a particular time and, therefore, who must pay for the shipping costs. The term **FOB** stands for *free on board* (or *freight on board*). When the vendor invoice specifies *FOB shipping point* (the most common business practice), legal title to the goods passes from the seller to the purchaser when inventory leaves the seller’s place of business. The purchaser, therefore, owns the goods while they are in transit and must pay the transportation costs. In the case of goods purchased *FOB shipping point*, the company purchasing the goods must include goods in transit from suppliers as units in inventory as of the year-end. In the case of goods purchased *FOB destination*, title to the goods does not pass from the seller to the purchaser until the goods arrive at the purchaser’s receiving dock. Therefore, these goods are not counted in year-end inventory of the purchasing company. Rather, the cost of these goods is included in inventory of the seller until the goods reach their destination.

This summary may help you understand the difference between *FOB shipping point* and *FOB destination*.

Shipping Terms	
FOB Shipping Point	FOB Destination
• Ownership of goods changes hands at the shipping point (i.e., the seller’s place of business) • The buyer pays for shipping • The buyer includes the goods in its inventory while the goods are in transit	• Ownership of goods changes hands at the destination (i.e., the buyer’s place of business) • The seller pays for shipping • The seller includes the goods in its inventory while the goods are in transit

Cost per Unit of Inventory. The cost per unit of inventory poses a challenge because companies purchase goods at different prices throughout the year. Which unit costs go into ending inventory? Which unit costs go to cost of goods sold?

For most inventory, it will be impossible to track the exact cost of items being sold and the exact cost of those that remain at the end of the financial year. Companies make use of inventory cost flow assumptions to help them determine cost of inventory. We will discuss these later, after we understand the basic inventory relationship.

Basic Inventory Relationship

The basic inventory principle will hold regardless of the inventory systems or cost assumptions you use (more on these two concepts later in this chapter). The model is extremely useful because it captures all inventory information for an entire accounting period. Study this model carefully (note that all amounts are assumed). Exhibit 6-4 presents this basic inventory principle.

Exhibit 6-4 Basic Inventory Relationship



This relationship reinforces what we already know about accounts, ending inventory, and cost of goods sold. If the beginning inventory is \$20 and we buy (or produce) \$100 worth of goods during a period, we can say that the total goods available for sale is \$120. Now, this does not mean that the entire \$120 worth of inventory is present all at once in a warehouse somewhere. There may be multiple purchases and sales during the period. Suppose we know that the ending inventory is \$30; we can say that cost of goods sold must have been \$90. Similarly, if we know cost of goods sold is \$90, we must have \$30 of inventory left on our Balance Sheet.

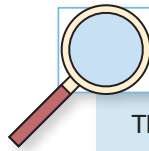
» Stop & Think

Zerillo & Sons started and ended the financial year with inventory totaling \$15,000 and \$28,000, respectively. Its cost of goods sold for the period totaled \$168,000. How much was the total inventory purchased during the period?

Answer:

$$\begin{aligned} \text{Beginning Inventory} + \text{Purchases} - \text{Ending Inventory} &= \text{Cost of Goods Sold} \\ \$15,000 + \text{Purchases} - \$28,000 &= \$168,000 \\ \text{So, Purchases} &= \$168,000 - \$15,000 + \$28,000 = \$181,000 \end{aligned}$$

You can rearrange the basic inventory relationship any way you wish in order to find the missing information.



A Closer Look

The cost of inventory on the Balance Sheet represents all the costs that the entity incurred to make its inventory ready for sale. *IAS 2—Inventories* states that the cost of inventories shall comprise all costs of purchase, costs of conversion, and other costs incurred in bringing the inventories to their present location and condition. The costs of purchase of inventories, thus, comprise the purchase price, import duties and other taxes, transport, handling, and other costs directly attributable to the acquisition of finished goods, materials, and services. If an entity receives trade discounts, rebates, and other similar items, they are deducted when determining the costs of purchase. Once the goods are ready for resale, other costs, such as advertising and sales commissions, are *not* included as the cost of inventory. Advertising, sales commissions, and delivery costs are selling expenses that go in the Income Statement, rather than in the Balance Sheet.



Stop & Think

During the month of June 20X4, Ten East, a shoe retailer, bought purchases totaling \$600,000. Ten East paid \$4,000 in delivery charges to get some shoes delivered to its store. Unfortunately, due to unsuitable goods or incorrect specifications, the company made purchase returns amounting to \$25,000. In addition, one of its suppliers also extended a purchase allowance of \$5,000, i.e., the supplier granted Ten East a subtraction from the amount owed. Ten East took advantage of early settlement discount schemes offered by its suppliers, and enjoyed a total of \$14,000 in discounts. What is Ten East's net cost of inventory purchases?

Purchase price of inventory	\$600,000
+ Freight-in (the cost to transport the goods from the seller to the buyer)	4,000
– Purchase returns for unsuitable goods returned to the seller	(25,000)
– Purchase allowances granted by the seller	(5,000)
– Purchase discounts for early payment by the buyer	(14,000)
= Net purchases of inventory (Cost to the buyer)	<u>\$560,000</u>

The next section shows how different accounting methods determine amounts reported on the Balance Sheet and the Income Statement. First, however, you need to understand how Inventory Accounting systems work.

RECORD INVENTORY-RELATED TRANSACTIONS

Inventory Systems



Record
inventory-related
transactions

There are two main types of Inventory Accounting systems: the periodic system and the perpetual system. The **periodic inventory system**, discussed in more detail in Appendix 6A, is typically used for inexpensive goods. A fabric store or a lumber yard won't keep a running record of every bolt of fabric or every piece of wood. Instead, these stores count their inventory periodically to determine the quantities on hand. Other businesses, such as restaurants and florists, may also use the periodic system because it does not make sense to keep track of the exact amount of meat, vegetables, cooking oil, and other ingredients that are used in producing the food items. This is an example of applying the “cost-benefit constraints” of the *Conceptual Framework*. The difference between quantity on hand and accounting records is simply considered used or sold.

A **perpetual inventory system**, on the other hand, tracks all inventory movements and typically uses a computer system to keep a running record of inventory on hand and what has been sold. This

system provides better information and control over inventory for companies such as FR, Carrefour, Nestlé, and a majority of other retailers. In fact, every time you see a retailer scanning bar codes of the products you are buying, it is more likely that the business is using a perpetual inventory system. The bar code on the product label holds lots of information. The optical scanner reads the bar code, and the computer system automatically records the sale and updates the inventory records.

Even with a perpetual system, a business must still count inventory on hand annually. The physical count establishes the correct amount of ending inventory for the financial statements and also serves as a check on the perpetual records. Any discrepancy may signal the possibility of inventory theft that we should be aware of. Here is a quick summary of the two main Inventory Accounting systems.

Perpetual Inventory System	Periodic Inventory System
• Used for all types of goods • Keeps a running record of all goods bought, sold, and on hand • Inventory counted at least once a year to determine any discrepancies between accounting records and inventory on hand	• Used for inexpensive goods • Does <i>not</i> keep a running record of all goods bought, sold, and on hand • Inventory counted at least once a year to determine ending inventory with any difference assumed to be used or sold

Recording Transactions in the Perpetual System

In a perpetual inventory system, all inventory transactions are tracked and accounting records are updated. At any point in time, the Inventory Account and cost of sales account are always up-to-date. Let's walk through a typical set of transactions related to the purchase and sale of inventory.

On January 1, 20X1, Natalie Quah started an online shop called We Love Baby Stuff to sell clothes for babies. She sources various manufacturers of baby clothing, orders select styles in bulk, and then sells the clothes to customers via her "blog shop."

Recording Inventory Purchases. Natalie selected and ordered a number of clothing items from a manufacturer. She was also able to negotiate a bulk **purchase discount** of 10% and a payment period of 30 days. When the goods arrived, they were accompanied by an invoice that also showed a total purchase price of \$2,000 (after deducting the bulk purchase discount) and a freight-inwards, delivery, or shipping charge of \$50. Freight-in is the transportation cost paid by the buyer to move goods from the seller to the buyer. Freight-in is accounted for as part of the cost of inventory, but some businesses may choose to keep track of it separately before adding it to the cost of inventory at a later point. Doing this will allow a business to track and monitor how much shipping charges it incurs. For example, Natalie could record either of the following journal entries:

A1				
	A	B	C	D
1	Jan 1			
2	20X1	Inventory	2,050	
3		Accounts Payable		2,050
4		Purchase of inventory, including \$50 freight costs		
5				

Option 1

A1				
	A	B	C	D
1	Jan 1			
2	20X1	Inventory	2,000	
3		Freight-inwards	50	
4		Accounts Payable		2,050
5		Purchase of inventory, including \$50 freight costs		
6				

Option 2

If Natalie uses Option 2, at the end of the accounting period, the freight-inwards should be transferred to the Inventory Account as part of the closing entry process (as discussed in Chapter 3).

A1				
	A	B	C	D
1	Jan 31			
2	20X1	Inventory	50	
3		Freight-inwards		50
4		<i>Transfer of freight-inwards to inventory cost</i>		
5				

Recording Purchase Returns. Sometimes, businesses may receive goods that fail to meet their specifications. For example, Natalie might have ordered 10 units of blue rompers, but green ones were delivered instead, or in the wrong sizes, or perhaps the items were defective. Depending on the purchase contract or terms of sales, Natalie may be able to ask for either a refund or a further discount. A **purchase return** is a decrease in the cost of inventory because the buyer returned the goods to the vendor. A **purchase allowance** also decreases the cost of inventory because the buyer got an allowance (a deduction) from the amount owed. To document approval of purchase returns, the vendor issues a **debit memorandum** (or debit memo), meaning that accounts payable are reduced (debited) for the amount of the return. The offsetting credit is to inventory as the goods are shipped back to the seller (vendor). Purchase discounts and allowances are usually documented on a revised invoice from the vendor. Suppose Natalie returned \$100 worth of products to her supplier and the vendor agreed to deduct this amount from the amount due. Natalie would record the following entry:

A1				
	A	B	C	D
1	Jan XX			
2	20X1	Accounts Payable	100	
3		Inventory		100
4		<i>Returned defective inventory to vendor</i>		
5				

The amount due to the vendor is now reduced by \$100 to \$1,950, called net purchases after deducting purchase returns and/or allowances. Throughout this book, we often refer to net purchases simply as *Purchases*.

Recording Inventory Sales. When Natalie's blog shop makes a sale, two things happen, in accordance with the *Conceptual Framework*.

- First, it now has an increase in assets, either in the form of cash or rights to receive cash (i.e., accounts receivable), which results in an increase in the business's net worth (via revenue).
- Second, it gave up some assets (i.e., inventory sold), which results in a reduction in the business's net worth (via expense).

For example, if Natalie sold a romper suit that costs \$15 for \$25, she would record the following journal entry:

A1				
	A	B	C	D
1	Jan XX			
2	20X1	Cash	25	
3		Sales Revenue		25
4		Cost of Sales	15	
5		Inventory		15
6		<i>Sold inventory costing \$15 for \$25 cash</i>		
7				

If Natalie incurs freight-out, delivery, or shipping costs in transporting her goods to the customers (i.e., under a FOB destination), such an expense is recorded as an operating expense, and not as part of cost of goods sold.

Recording Settlement Discount. We mentioned earlier that Natalie purchased her inventory from suppliers on credit. Usually, you may see the term “net 30,” which tells the buyer to pay the full amount within 30 days. Occasionally, you may see payment terms such as “2/10, n/30.” This means the buyer can take a 2% discount for early payments made within 10 days, otherwise the invoiced amount is due within 30 days. The discount is applied to the amount owed from goods purchased and excludes other charges such as shipping and handling costs. Of course, other suppliers may offer different payment terms, such as 1/15, n/45.

Suppose Natalie ordered another line of baby clothing from a different vendor, totaling \$1,000. This particular vendor offers a “2/10, n/30” with free shipping and handling. Upon receipt of inventory from the supplier, Natalie recorded the following.

A1				
	A	B	C	D
1	Jan XX			
2	20X1	Inventory	1,000	
3		Accounts Payable		1,000
4		<i>Purchase of inventory on 2/10, n/30 terms</i>		
5				

If Natalie decides to take advantage of the early settlement discount, she would pay \$980 to settle the \$1,000 obligation with the following journal entry.

A1				
	A	B	C	D
1	Jan XX			
2	20X1	Accounts Payable	1,000	
3		Cash		980
4		Discounts Received (or Inventory)		20
5		<i>Payment for inventory with 2/10 settlement discount</i>		
6				

Discount received is a reduction in the cost of inventory purchased. Similar to our earlier discussion on freight-inwards, if Natalie wants to keep track of the various settlement discounts she has received, she can use a specific account to keep track of it. At the end of the accounting period, discount received is closed to the Inventory Account. Alternatively, she can reduce the cost of inventory directly. Naturally, if Natalie does not act on the early settlement discount, the account payable will be paid in full by cash (\$1,000) when payment is made.

UNDERSTAND AND APPLY DIFFERENT INVENTORY COST ASSUMPTIONS

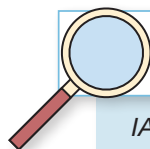
Inventory is an example of an asset for which a manager can decide which accounting method to use. The accounting method selected affects the profits to be reported, the amount of income tax to be paid, and the values of the financial ratios derived from the financial statements.

Determining the cost of inventory is easy when the unit cost remains constant, as in Exhibit 6-2. But the unit cost usually changes. For example, prices often rise. The trousers that cost UNIQLO \$10 in January may cost \$14 in June and \$18 in October. Suppose UNIQLO sells 1,000 pairs of trousers in November. How many of those cost \$10, how many cost \$14, and how many cost \$18?

To compute cost of goods sold and the cost of ending inventory still on hand, we must assign unit cost to the items. *IAS 2—Inventories* prescribes that cost of inventory items that are not ordinarily interchangeable (and goods or services produced and segregated for specific projects) shall

3 Understand and apply different inventory cost assumptions

be assigned by using **specific identification** of their individual costs. Cost of other inventories that are ordinarily interchangeable shall be determined using **cost formulas**. The common cost formulas (also known as **inventory costing methods**) include **first-in, first-out (FIFO)**; **last-in, first-out (LIFO)**; and **weighted average cost method**.



A Closer Look

IAS 2—Inventories prohibits the use of the last-in, first-out (LIFO) cost formula. This is one of the major differences between the U.S. and International GAAP. For the purpose of this chapter, we will continue to elaborate on LIFO because it provides a good contrast to FIFO, one of the cost formulas allowed under IFRS.

IAS 2—Inventories also allows the *retail inventory method* to be used, as long as it approximates the cost. This method is used by certain retailers for measuring inventories of large numbers of rapidly changing items with similar margins. The cost of inventory is determined by reducing the sales value of the inventory by the appropriate percentage gross margin. The percentage used takes into consideration inventory that has been marked down to below its original selling price. As this is an industry-specific cost formula, we will not be discussing this method in detail. If you want to find out more, you can visit <https://www.accountingtools.com/articles/2017/5/13/retail-inventory-method>.



The inventory costing methods can have very different effects on reported profits in the Income Statement and inventory on the Balance Sheet. Therefore, companies select their inventory method with great care. *IAS 2* requires an entity to use the same cost formula for all inventories having a similar nature and use to the entity. It is possible to use different cost formulas for inventories with a different nature or use. For example, Chow Tai Fook, a leading jeweler in Asia, uses different inventory methods for different inventory items.

ADAPTED EXCERPTS FROM CHOW TAI FOOK'S NOTES TO THE ACCOUNTS

Inventories

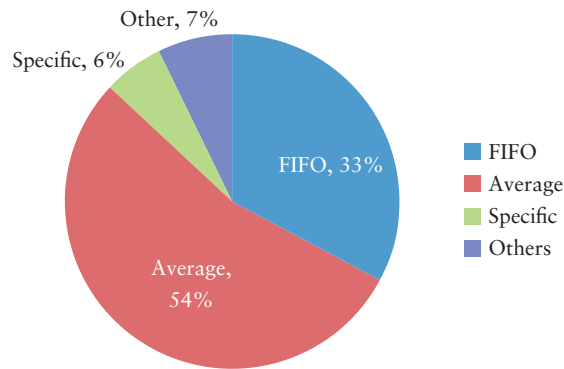
Inventories are stated at the lower of cost and net realizable value. Cost is calculated using *specific identification* basis for gem-set jewelry and watches, and weighted average for other inventories.

Source: From Chow Tai Fook, Annual Report, 2016, page 154.

Exhibit 6-5, based on a survey of 170 IFRS companies in 2010, indicates that weighted average is the most popular inventory costing method. The same survey also shows that 30 out of the 170 companies employ more than one inventory costing method, like Chow Tai Fook.

Let's start with looking at Exhibit 6-6, a diagrammatic overview of how specific identification and the other cost formulas work. Boxes 1, 2, 3, and 4 are units purchased at different prices in a chronological order. For now, just look at how the different inventory costing methods will result in different costs being reported on the Balance Sheet as ending inventory and on the Income Statement as cost of goods sold.

Specific Identification. Some businesses deal in unique inventory items, such as automobiles, antique furniture, jewelry, and real estate. These businesses cost their inventories at the specific cost of the particular unit. For instance, a Toyota dealer may have four vehicles in the showroom—(1) a Camry, (2) a Corolla, (3) a Prius, and (4) a Lexus. If the dealer sells the Camry, Prius, and Lexus, the cost of goods sold is the cost of the three vehicles (1, 3, and 4) sold. The only item remaining on its Balance Sheet will be the (2) Corolla.

Exhibit 6-5 Use of the Various Inventory Methods

Source: Exhibit created from source data found in IFRS Financial Statements – Best Practices in Presentation and Disclosure, 2012/13, AICPA.

The specific identification method is also called the specific-unit-cost method. For most businesses, this method is too expensive to use for inventory items that have common characteristics, such as bushels of wheat, gallons of paint, or the casual apparel that UNIQLO sells.

The other inventory costing methods, or cost formulas (FIFO, LIFO, and weighted average), are fundamentally different. These other methods do not use the specific cost of a particular unit. Instead, they assume different flows of inventory costs. Remember that the goods are interchangeable, and cost formulas are not the same as the physical units. Businesses will always try to sell older units first (to prevent spoilage, obsolescence, etc.) before selling newer units. This is independent of the cost formulas used.

To illustrate FIFO, LIFO, and average inventory costing methods, we use a common set of data, given in Exhibit 6-7.

For Exhibit 6-7, let's assume UNIQLO began the period with 10 khakis that cost \$10 each; thus, the beginning inventory was \$100. During the period, UNIQLO bought 50 more khakis (at different prices), sold 40 khakis, and ended the period with 20 khakis. These have been summarized in Exhibit 6-7.

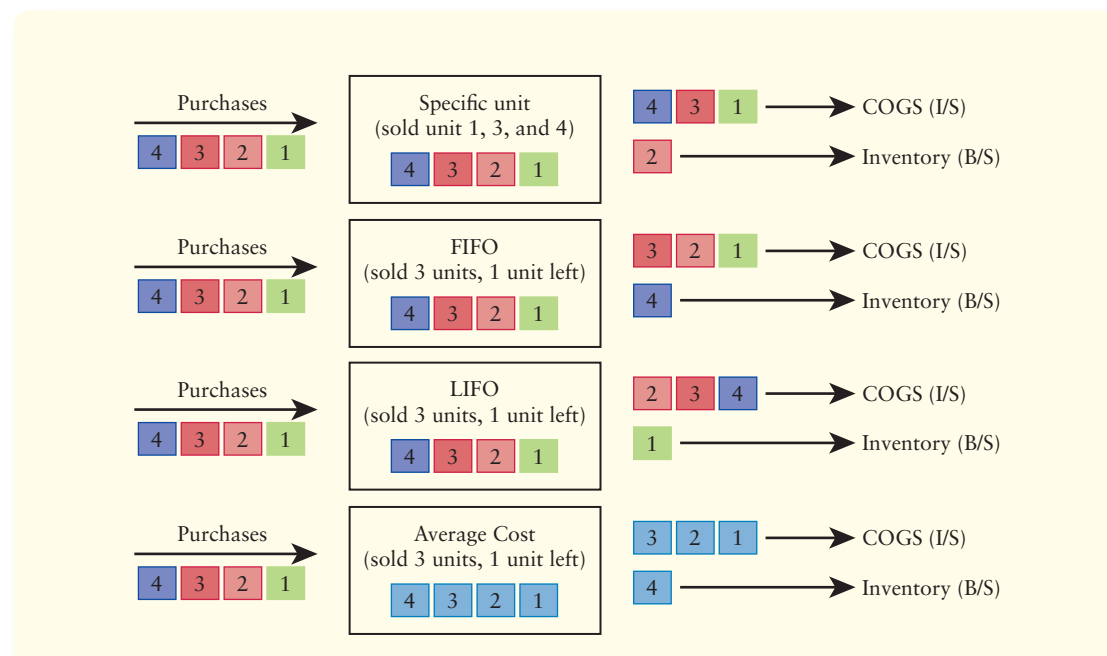
Exhibit 6-6 Overview of Costing Methods

Exhibit 6-7 Inventory Data Used to Illustrate the Various Inventory Costing Methods

Inventory			
Beg. bal.	(10 units @ \$10)	100	
Purchases:			Cost of goods sold
No. 1	(25 units @ \$14)	350	(40 units @ ?)
No. 2	(25 units @ \$18)	450	
End. bal.	(20 units @ ?)	?	

Goods Available		Number of Units	Total Cost
Goods available	=	10 + 25 + 25 = 60 units	\$100 + \$350 + \$450 = \$900
Cost of goods sold	=	40 units	?
Ending inventory	=	20 units	?

The big accounting questions are:

1. What is the cost of goods sold for the Income Statement?
2. What is the cost of the ending inventory for the Balance Sheet?

It all depends on which inventory method UNIQLO uses. Let's look at FIFO costing first.

FIFO Cost. Under the FIFO method, the first costs into inventory are the first costs assigned to cost of goods sold—hence, the name *first-in, first-out*. The following T-account shows how to compute FIFO cost of goods sold and ending inventory for the UNIQLO khakis (data from Exhibit 6-7):

Inventory (at FIFO cost)			
Beg. bal.	(10 units @ \$10)	100	
Purchases:			Cost of goods sold (40 units):
No. 1	(25 units @ \$14)	350	(10 units @ \$10) 100
No. 2	(25 units @ \$18)	450	(25 units @ \$14) 350
			(5 units @ \$18) 90
End. bal.	(20 units @ \$18)	360	540

Under FIFO, the cost of ending inventory is always based on the latest costs incurred—in this case \$18 per unit.

LIFO Cost. LIFO costing is the opposite of FIFO. Under LIFO (*last-in-first-out*), the last costs into inventory go immediately to cost of goods sold. The following T-account shows how to compute the LIFO inventory amounts for the UNIQLO khakis (data from Exhibit 6-7):

Inventory (at LIFO cost)			
Beg. bal.	(10 units @ \$10)	100	
Purchases:			Cost of goods sold (40 units):
No. 1	(25 units @ \$14)	350	(25 units @ \$18) 450
No. 2	(25 units @ \$18)	450	(15 units @ \$14) 210
End. bal.	(10 units @ \$10)		
	(10 units @ \$14)	240	660

Under LIFO, the cost of ending inventory is always based on the oldest costs—from beginning inventory plus the early purchases of the period—\$10 and \$14 per unit.

Average Cost. The *average-cost method*, sometimes called the weighted-average method, is based on the average cost of inventory during the period. Unlike LIFO and FIFO, the individual

inventory costs are no longer relevant when inventory is sold. Average cost per unit is determined as follows (data from Exhibit 6-7):

$$\text{Average cost per unit} = \frac{\text{Cost of goods available}^*}{\text{Number of units available}^*} = \frac{\$900}{60} = \$15$$

$$\begin{aligned} \text{Cost of goods sold} &= \text{Number of units sold} \times \text{Average cost per unit} \\ &= 40 \text{ units} \times \$15 = \$600 \end{aligned}$$

$$\begin{aligned} \text{Ending inventory} &= \text{Number of units on hand} \times \text{Average cost per unit} \\ &= 20 \text{ units} \times \$15 = \$300 \end{aligned}$$

The following T-account shows the effects of average costing:

Inventory (at Average Cost)			
Beg. bal.	(10 units @ \$10)	100	
Purchases:			
No. 1	(25 units @ \$14)	350	
No. 2	(25 units @ \$18)	450	
			Cost of goods sold (40 units @ average cost of \$15 per unit) 600
End. bal.	(20 units @ average cost of \$15 per unit)	300	

Effects of FIFO, LIFO, and Average Cost on Cost of Goods Sold, Gross Profit, and Ending Inventory

In our UNIQLO trousers example, the cost of inventory rose from \$10 to \$14 to \$18. When inventory unit costs change this way, the various inventory methods produce different cost-of-goods sold figures. Exhibit 6-8 summarizes the income effects (sales – cost of goods sold = gross profit) of the three inventory methods (remember that prices are rising). Study Exhibit 6-8 carefully, focusing on cost of goods sold and gross profit.

Exhibit 6-8 Financial Statement Effects of the FIFO, LIFO, and Average Cost Inventory Methods

	FIFO	LIFO	Average
Sales revenue (assumed)	\$1,000	\$1,000	\$1,000
Cost of goods sold.....	<u>540</u> (lowest)	<u>660</u> (highest)	<u>600</u>
Gross profit.....	<u>\$ 460</u> (highest)	<u>\$ 340</u> (lowest)	<u>\$ 400</u>
Inventory (on Balance Sheet)	\$ 360 (highest)	\$ 240 (lowest)	\$ 300

Exhibit 6-9 shows the impact of both FIFO and LIFO costing methods on cost of goods sold and inventories during both increasing costs (Panel A) and decreasing costs (Panel B). Study this exhibit carefully; it will help you *really* understand FIFO and LIFO.

*Goods available = Beginning inventory + Purchases

Exhibit 6-9 Cost of Goods Sold and Ending Inventory: FIFO and LIFO; Increasing and Decreasing Costs

Panel A—When Inventory Costs Are Increasing

	Cost of Goods Sold (COGS)	Ending Inventory
FIFO	FIFO COGS is lowest because it's based on the oldest costs, which are low. Gross profit is, therefore, the highest.	FIFO ending inventory is highest because it's based on the most recent costs, which are high.
LIFO	LIFO COGS is highest because it's based on the most recent costs, which are high. Gross profit is, therefore, the lowest.	LIFO ending inventory is lowest because it's based on the oldest costs, which are low.

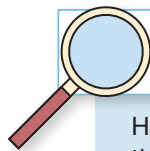
Panel B—When Inventory Costs Are Decreasing

	Cost of Goods Sold (COGS)	Ending Inventory
FIFO	FIFO COGS is highest because it's based on the oldest costs, which are high. Gross profit is, therefore, the lowest.	FIFO ending inventory is lowest because it's based on the most recent costs, which are low.
LIFO	LIFO COGS is lowest because it's based on the most recent costs, which are low. Gross profit is, therefore, the highest.	LIFO ending inventory is highest because it's based on the oldest costs, which are high.

Comparison of the Inventory Methods

Let's compare the average, FIFO, and LIFO inventory methods.

1. Measuring Cost of Goods Sold. How well does each method match inventory expense—cost of goods sold—against revenue? LIFO results in the most realistic net income figure because LIFO assigns the most recent inventory costs to expense. In contrast, FIFO matches older inventory costs against revenue—a poor measure of expense. FIFO income is, therefore, less realistic than LIFO income.
2. Measuring Ending Inventory. Which method reports the most up-to-date inventory cost on the Balance Sheet? FIFO. LIFO can value inventory at very old costs because LIFO leaves the oldest prices in ending inventory.

**A Closer Look**

How would the method of recording inventory transactions and cost flow assumptions interact with each other? The frequency of recording inventory transactions (perpetual versus periodic) affects how we calculate weighted average and LIFO. For example, let's consider the following simple illustration.

Suppose Natalie's We Love Baby Stuff has an opening inventory balance of \$1,000 (100 units) on July 1, 20X2. Natalie sold 30 units of inventory for \$400 on July 10 and bought an additional 50 units of inventory totaling \$575 on July 20. Under the "periodic average," the average inventory cost is \$10.50 ($\$1,000 + \575 for 150 units), resulting in ending inventory of \$1,260 (120 units $\times$ \$10.50) and cost of sales of \$315 (30 units $\times$ \$10.50).

However, if a "perpetual average" is used, we will calculate a new weighted average cost every time we make a purchase. In this example, at the time of the sale on July 10, the average cost per inventory item was still \$10 (based on 100 units of opening inventory for \$1,000), so the cost of sales is \$300 (30 units $\times$ \$10 per unit). After the additional inventory purchase on July 20, the "new" average cost per unit is now \$10.63 ($\$700 + \575 for 120 units). Any sales Natalie makes after July 20 will use this \$10.63 as the average cost (until new purchases that will produce a newer average cost). A similar concept applies to "periodic LIFO" versus "perpetual LIFO."

As perpetual inventory systems use computerized accounting systems, these calculations are usually automated. Unless you are given specific dates and quantities of inventory purchases and sales, you will not be able to work out the perpetual variations of LIFO or weighted average. In such cases, for simplicity, you can assume that the cost of sales and ending inventory are calculated at the end of the period.

Mid-Chapter

Summary Problem

Suppose a division of HH Precision Industry that sells computer microchips has these inventory records for January 20X7:

Date	Item	Quantity	Unit Cost	Total Cost
Jan 1	Beginning inventory	100 units	\$ 8	\$ 800
6	Purchase	60 units	9	540
21	Purchase	150 units	9	1,350
27	Purchase	90 units	10	900

Company accounting records show sales of 310 units for revenue of \$6,770. Operating expense for January was \$1,900.

Requirements

- Prepare the January Income Statement, showing the amounts for LIFO, average, and FIFO cost. Label the bottom line "Operating income." Round the average cost per unit to three decimal places and all other figures to whole-dollar amounts. Show your computations.
- Suppose you are the financial vice president of HH Precision Industry. Which inventory method will you use if your motive is to
 - minimize income taxes?
 - report the highest operating income?

- c. report operating income between the extremes of FIFO and LIFO?
- d. report inventory on the Balance Sheet at the most current cost?
- e. attain the best measure of net income for the Income Statement?

State the reason for each of your answers.

Answers

Requirement 1

	A	B	C	D
1	HH Precision Industry			
2	Income Statement for Microchip			
3	Month Ended January 31, 20X7			
4		LIFO	Average	FIFO
5	Sales revenue	\$ 6,770	\$ 6,770	\$ 6,770
6	Cost of goods sold	2,870	2,782	2,690
7	Gross profit	3,900	3,988	4,080
8	Operating expenses	1,900	1,900	1,900
9	Operating income	\$ 2,000	\$ 2,088	\$ 2,180

Cost of goods sold computations:

$$\text{LIFO: } (90 @ \$10) + (150 @ \$9) + (60 @ \$9) + (10 @ \$8) = \$2,870$$

$$\text{Average: } 310 \times \$8.975^* = \$2,782$$

$$\text{FIFO: } (100 @ \$8) + (60 @ \$9) + (150 @ \$9) = \$2,690$$

$$^* \frac{(\$800 + \$540 + \$1,350 + \$900)}{(100 + 60 + 150 + 90)} = \$8.975$$

Requirement 2

- a. Use LIFO to minimize income taxes. Operating income under LIFO is lowest when inventory unit costs are increasing, as they are in this case (from \$8 to \$10). (If inventory costs were decreasing, income under FIFO would be lowest.)
- b. Use FIFO to report the highest operating income. Income under FIFO is highest when inventory unit costs are increasing, as in this situation.
- c. Use the **average-cost method** to report an operating income amount between the FIFO and LIFO extremes. This is true in this situation and in others when inventory unit costs are increasing or decreasing.
- d. Use FIFO to report inventory on the Balance Sheet at the most current cost. The oldest inventory costs are expensed as cost of goods sold, leaving in ending inventory the most recent (most current) costs of the period.
- e. Use LIFO to attain the best measure of net income. LIFO produces the best matching of current expense with current revenue. The most recent (most current) inventory costs are expensed as cost of goods sold.

Other Inventory Issues

Comparability as an Enhancing Qualitative Characteristic. The **comparability** qualitative characteristic states that businesses should use the same accounting methods and procedures from period to period. Consistency enables investors to compare a company's financial statements from one period to the next.

Suppose you are analyzing a company's net income pattern, which showed an increase over a two-year period. However, you also noted that the company switched from LIFO to FIFO during that time. Its net income increased dramatically but only because of the change in inventory method. If you did not know of the accounting change, you might believe that the company's income increased due to improved operations, but that's not the case.

The comparability qualitative characteristic does not mean that a company is not permitted to change its accounting methods. As long as the change reflects a more relevant and faithful representation of an underlying economic phenomenon, the lack of (short-term) comparability should not prevent the accounting method change. Typically, a reporting entity that makes an accounting change must disclose the effect of the change on net income or restate the previous year's figures for comparative purposes.

For example, HollyFrontier Corporation changed its inventory cost method from FIFO to LIFO in 2009. In its press release on the accounting policy change, HollyFrontier stated that its pre-tax income would decrease by \$180 million.

On the other hand, Pactiv Corporation changed its inventory cost method from a combination of LIFO and FIFO to exclusively FIFO. It claimed that this change would better reflect the current value of inventories, provide better matching of sales and expenses, provide uniformity across all operations with respect to the method of Inventory Accounting, and enhance comparability with peers. Another reason cited for the switch was that the convergence of U.S. and International GAAP would likely eliminate LIFO because IFRS does not allow the use of LIFO.

Clearly, the choice of inventory method is an important one for all companies that deal with inventory.

Why is LIFO not allowed under IFRS? In the introduction section of *IAS 2—Inventories*, IASB simply states: "The Standard does not permit the use of the last-in, first-out (LIFO) formula to measure the cost of inventories." So why does IASB prohibit the use of LIFO?

We said earlier that using the latest costs is a better measure of income, and, on the other hand, using older costs results in more appropriate inventory values on the Balance Sheet. The *Conceptual Framework* favors a Balance Sheet primacy. Changes in assets and liabilities determine income and net worth of a business. LIFO prioritizes income measurements over that of assets and liabilities and results in inventories being recognized on the Balance Sheet at amounts that bear little relationship to cost level of inventories. In its basis for conclusions for *IAS 2*, the IASB stated: "It is not appropriate to allow an approach that results in a measurement of profit or loss for the period that is inconsistent with the measurement of inventories for Balance Sheet purposes."

In fact, LIFO can also distort income when inventory is depleted to a very low level, as much older costs (that could have been in the inventory for years) are used, creating unrealistic profits. Furthermore, IASB explained that the concept of selling newer items first before older ones is "generally not a reliable representation of actual inventory flows," and that LIFO lacks "representation faithfulness of inventory flows."

What is interesting is how the accounting standard is determined at the highest level. The IASB uses the *Conceptual Framework* to determine what should be the most appropriate accounting treatment for items on the financial statements.

Net Realizable Value. All our previous discussions have focused on the cost aspect of inventory. Regardless of which cost method you are using, there is one additional aspect that is equally important. *IAS 2* requires inventories to be measured at the lower of cost and net realizable value

(NRV). In other words, once you have determined cost (using specific identification, FIFO, or average cost method), you will need to compare the inventory to its NRV. The lower of the two shall be what is reported on the Balance Sheet. **Net realizable value** is the estimated selling price in the ordinary course of business less the estimated costs of completion and the estimated costs necessary to make the sale.

This evaluation of cost versus NRV is necessary because an entity may not be able to recover the cost of inventory if the goods are damaged or obsolete, or if their selling price has declined below costs. If an inventory is written down to an NRV below cost, the write-down is recognized as an expense in the period of the write-down. Any subsequent reversal is recognized as a reduction in COGS during the period in which the reversal occurs. The exact determination of NRV is not within the scope of this textbook, but the example below demonstrates the concept of lower of cost and NRV.

Suppose UNIQLO paid \$3,000 for inventory on September 26. By January 31, its financial year-end, \$2,000 of the inventory has been sold, and the cost of the remaining inventory is \$1,000. Let's further assume that some of these inventories are damaged and no longer suitable for normal sale. UNIQLO estimates the net realizable value of these goods to be \$400. UNIQLO's year-end Balance Sheet must report this inventory at NRV of \$400. Exhibit 6-10 presents the effects of NRV on the Balance Sheet and the Income Statement. Before any NRV effect, cost of goods sold is \$2,000. An NRV write-down decreases inventory and increases cost of goods sold, as follows:

A1		
	A	B C
1	Cost of Goods Sold	600
2	Inventory	600
3	Wrote inventory down to realizable value.	
4		

If the NRV of UNIQLO's inventory had been above cost, it would have made no adjustment for NRV. In that case, simply report inventory at cost, which is the lower of cost and NRV.

Exhibit 6-10 | Net Realizable Value (NRV) Effects on Inventory and Cost of Goods Sold

Balance Sheet

Current assets:

Cash	\$ XXX
Short-term investments	XXX
Accounts receivable.....	XXX
Inventories, net realizable value	
(which is lower than \$1,000 cost)	400
Prepaid expenses	XXX
Total current assets	\$X,XXX

Income Statement

Sales revenue	\$ XXX
Cost of goods sold (\$2,000 + \$600)	2,600
Gross profit	\$X,XXX

Let's have a look at how Bossini, another clothing retailer, provides disclosure on its determination of NRV (adapted).

ADAPTED EXCERPTS FROM BOSSINI'S NOTES TO THE ACCOUNTS

Inventories

Management reviews the aged analysis of inventories of the Group at the end of each reporting period, and makes allowance for obsolete and slow-moving inventory items identified that are no longer suitable for sale. The assessment of allowance amount required involves management judgements and estimates. When the actual outcome or expectation in future is different from the original estimate, such differences will have an impact on the carrying value of the inventories and allowance charge/write-back of allowance in the period in which such estimate has been changed. In addition, physical counts on all inventories are carried out on a periodical basis in order to determine whether allowance needs to be made in respect of any obsolete inventories identified. The Group carries out an inventory review at the end of each reporting period and makes provision against obsolete and slow-moving items. Management reassesses the estimation at the end of each reporting period. The Group is satisfied that sufficient allowance for obsolete and slow-moving inventories has been made in the consolidated financial statements.

Source: From Bossini, Annual Report, 2015, page 112–119.

As mentioned in the excerpt, Bossini uses an allowance for inventory obsolescence account. The concept is similar to the allowance for uncollectible account we saw in Chapter 5 for receivables.

Let's take another look at FR's inventory excerpt that we saw at the start of this chapter.

ADAPTED EXCERPTS FROM FAST RETAILING'S NOTES TO THE ACCOUNTS

Inventories

Inventories are valued at the lower of cost and net realizable value; the weighted average method is principally used to determine cost. Net realizable value is based on the estimated selling price in the ordinary course of business less any estimated costs to be incurred to sell the goods.

Source: From Fast Retailing, Annual Report, 2015 page 10.

You should have a better understanding of the policy now. Notice FR is using the weighted average method, and it makes sure inventory is valued at the lower of cost and net realizable value.

ANALYZE EFFECTS OF INVENTORY ERRORS

Inventory errors sometimes occur. An error in ending inventory creates errors for two accounting periods. For example, if ending inventory has been *overstated* by \$5,000, cost of goods sold will thus be *understated* by the same amount, resulting in an *overstated* gross profit. In addition, the error in ending inventory in period 1 will also impact the beginning inventory in period 2. In the absence of any further errors, inventory errors counterbalance in two consecutive periods. Why? Period 1's ending inventory becomes period 2's beginning amount. Thus, the period 1 error carries over into period 2, resulting in an *overstated* cost of goods sold, and an *understated* gross profit in period 2.

In Exhibit 6-11, we show the effect of the above inventory errors over a three-year period, with constant sales and purchases. Had there been no error in period 1, the figures under period 3 are the correct ones. You can compare the figures in period 1 and 2 to those in period 3 for a better understanding of the impact of the inventory errors.

Analyze effects of inventory errors

Exhibit 6-11 Inventory Errors: An Example

A1						
	A	B	C	D	E	F
		Period 1		Period 2		Period 3
		Ending Inventory Overstated by \$5,000		Beginning Inventory Overstated by \$5,000		Correct
1						
2	Sales revenue		\$ 100,000		\$ 100,000	\$ 100,000
3	Cost of goods sold:					
4	Beginning inventory	\$ 10,000		\$ 15,000		\$ 10,000
5	Purchases	50,000		50,000		50,000
6	Cost of goods available	60,000		65,000		60,000
7	Ending inventory	(15,000)		(10,000)		(10,000)
8	Cost of goods sold		45,000		55,000	50,000
9	Gross profit		\$ 55,000		\$ 45,000	\$ 50,000
10						
11				100,000		
12						

The authors thank Professor Carl High for this example.

Beginning inventory and ending inventory have opposite effects on cost of goods sold (beginning inventory is added; ending inventory is subtracted). Therefore, after two periods, an inventory error washes out (counterbalances). Notice that total gross profit is correct for periods 1 and 2 combined (\$100,000) even though each year's gross profit is off by \$5,000. The correct gross profit is \$50,000 for each period, as shown in period 3.

We must have accurate information for all periods. Exhibit 6-12 summarizes the effects of Inventory Accounting errors.

Exhibit 6-12 Effects of Inventory Errors

A1					
	A	B	C	D	E
		Period 1		Period 2	
		Cost of Goods Sold	Gross Profit and Net Income	Cost of Goods Sold	Gross Profit and Net Income
1	Inventory Error				
2	Period 1				
3	Ending inventory overstated	Understated	Overstated	Overstated	Understated
4	Period 1				
5	Ending inventory understated	Overstated	Understated	Understated	Overstated
6					



COOKING THE BOOKS
with Inventory

Crazy Eddie

It is one thing to make honest mistakes in accounting for inventory, but quite another to use inventory to commit fraud. The two most common ways to “cook the books” with inventory are as follows:

1. Inserting fictitious inventory, thus overstating quantities
2. Deliberately overstating unit prices used in the computation of ending inventory amounts

Either one of these tricks has exactly the same effect on income as inventory errors, discussed in the previous section. The difference is that honest inventory errors are often corrected

as soon as they are detected, thus minimizing their impact on income. In contrast, deliberate overstatement of inventories tends to be repeated over and over again throughout the course of months, or even years, thus causing the misstatement to grow ever higher until it is discovered. By that time, it can be too late for the company.

Crazy Eddie, Inc.,¹ was a retail consumer electronics store in 1987, operating 43 retail outlets in the New York City area, with \$350 million in reported sales and reported profits of \$10.5 million. Its stock was a Wall Street “darling,” with a collective market value of \$600 million. The only problem was that the company’s reported profits had been grossly overstated since 1984, the year that the company went public.

Eddie Antar, the company’s founder and major shareholder, became preoccupied with the price of his company’s shares in 1984. Antar realized that the company, in an extremely competitive retail market in the largest city in the United States, had to keep posting impressive operating profits in order to maintain the upward trend in the company’s share price.

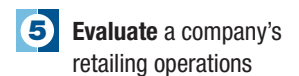
Within the first six months, Antar ordered a subordinate to double count about \$2 million of inventory in the company’s stores and warehouses. Using Exhibits 6-14 and 6-15, you can see that the impact of this inventory overstatement went straight to the “bottom line,” overstating profits by the same amount. Unfortunately, the company’s auditors failed to detect the inventory overstatement. The following year, emboldened by the audit error, Antar ordered subordinates (now accomplices) to bump the overstatement to \$9 million. In addition, he ordered employees to destroy incriminating documents to conceal the inventory shortage. When auditors asked for these documents, employees told them they had been lost. Antar also ordered that the company scrap its sophisticated computerized perpetual inventory system and return to an outdated manual system that was easier to manipulate. The auditors made the mistake of telling Antar which company stores and warehouses they were going to visit in order to observe the year-end physical count of inventory. Antar shifted sufficient inventory to those locations just before the counts to conceal the shortages. By 1988, when the fraud was discovered, the inventory shortage (overstatement) was larger than the total profits the company had reported since it went public in 1984.

In June 1989, Crazy Eddie, Inc., filed for Chapter 11 bankruptcy protection. Later that year, the company closed its stores and sold off its assets. Eddie Antar became a fugitive from justice, moved to Israel, and took an assumed name. He was arrested in 1992, extradited to the United States, and convicted on 17 counts of fraudulent financial reporting in 1993. He was ordered to pay \$121 million in restitution to former shareholders and creditors.

A series of missteps by the courts led to a plea bargain agreement in 1996, a condition of which Antar admitted, for the first time, that he had defrauded investors by manipulating the company’s accounting records. One of the prosecuting attorneys was quoted as saying, “Crazy Eddie wasn’t crazy, just crooked.”

EVALUATE A COMPANY’S RETAILING OPERATIONS

Owners, managers, and investors use ratios to evaluate a business. Two ratios relate directly to inventory: gross profit percentage and the rate of inventory turnover.



Gross Profit Percentage. Gross profit (sales — cost of goods sold) is a key indicator of a company’s ability to sell inventory at a profit. Merchandisers strive to increase **gross profit percentage**, also called the **gross margin percentage**. Gross profit percentage is mark-up stated as a percentage of sales. Gross profit percentage is computed as follows for FR. Data (in billions of ¥) for 2015 was taken from FR’s financial statements at the start of this chapter.

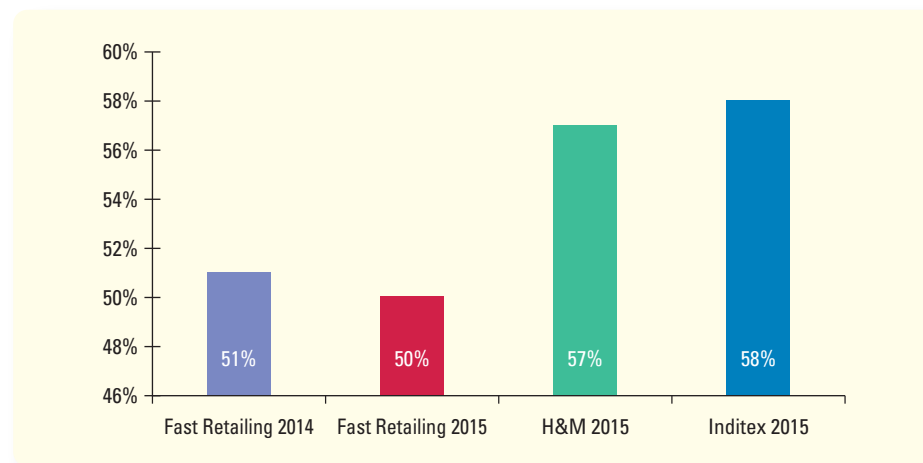
Gross Profit Margin	Fast Retailing 2015	Fast Retailing 2014
$\frac{\text{Gross profit}}{\text{Sales}}$	$\frac{848.6}{1,681.8} = 51\%$	$\frac{699.7}{1,382.9} = 51\%$

¹ Michael C. Knapp, *Contemporary Auditing: Real Issues and Cases*, 6th edition, Mason, Ohio: Thomson Southwestern, 2009.

The gross profit percentage is watched carefully by managers and investors. A 50% gross margin means that each dollar of sales generates 50 cents of gross profit. In other words, cost of goods sold consumes 50 cents of each sales dollar for FR. You can see that FR's gross profit margin has declined slightly from 51% the year before. Many firms do not enjoy such a high gross profit margin as FR. But the top apparel retailers are doing well. Inditex (Zara and other brands) and H&M have gross profit margins of 58% and 57%, respectively.

You may also find that fashion retailers generally have higher gross profit margins than, say, supermarkets and department stores. Carrefour, for example, usually has gross profit margins of around 20%. For most merchandising firms, the gross profit percentage changes little from year to year, but a small change may have a big impact on the bottom line. Exhibit 6-13 graphs the gross profit percentages for these companies. We will further discuss how to evaluate a company's performance in comparison with its peers later (in Chapter 12).

Exhibit 6-13 Gross Profit Percentages of Selected Retailers



Inventory Turnover and Resident Period. Fast Retailing strives to sell its inventory as quickly as possible because the goods generate no profit until they're sold. The faster the sales, the higher the income; on the other hand, slow-moving goods generate fewer sales and lower income. Ideally, a business should operate with as little inventory as possible, but most businesses, especially retailers, must keep some goods on hand. **Inventory turnover**, the ratio of cost of goods sold to average inventory, indicates how rapidly inventory is sold. The 2015 computation for FR follows (data in billions of ¥) from FR's financial statements at the start of this chapter:

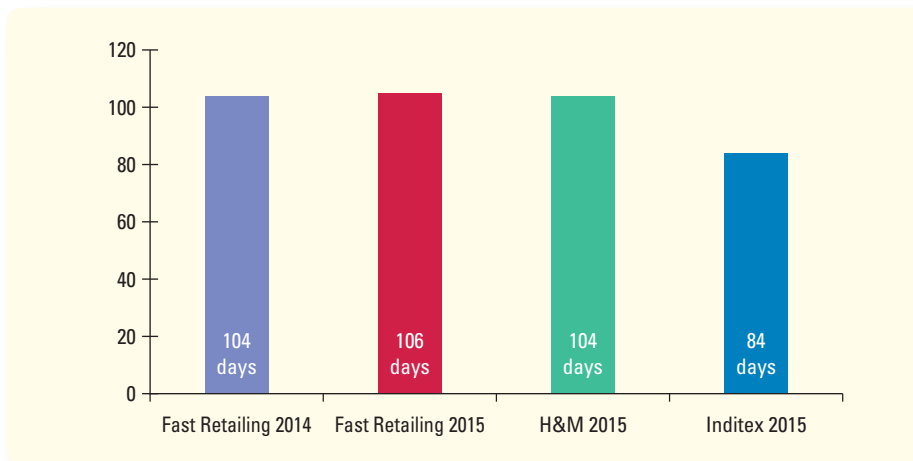
Inventory Turnover and Resident Period	Fast Retailing 2015	Fast Retailing 2014
$\frac{\text{Cost of goods sold}}{\text{Average inventory}^*}$	$\frac{833.2}{241.6} = 3.45 \text{ times}$	$\frac{683.2}{195.4} = 3.50 \text{ times}$
365 / Inventory turnover	$365 / 3.45 = 106 \text{ days}$	$365 / 3.50 = 104 \text{ days}$

*Average inventory = (beginning balance + ending balance)/2, which is (260 + 223.2)/2 = 241.6 for 2015

The inventory turnover ratio shows how many times the company sold (or turned over) its average level of inventory during the year. The turnover can also be expressed in days, called **inventory resident period** or days inventory on hand, by dividing 365 by the turnover ($365/3.45 = 106$ days). Inventory turnover varies from industry to industry.

Exhibit 6-14 charts the rates of inventory turnover for the top three apparel retail companies in the world. You saw earlier that FR's inventory days were 106 days in 2015 and 105 days in 2014. Inditex and H&M inventory days are 104 days and 84 days, respectively. You can see that Inditex has the highest gross margin and has the fastest inventory turnover. That's one reason Inditex is the top apparel company in the world, at least in financial performance terms.

Exhibit 6-14 Inventory Resident Period of Selected Retailers



» Stop & Think

Beginning inventory is \$70,000, net purchases total \$365,000, and net sales are \$500,000. With a normal gross profit ratio of 40% of sales (i.e., cost of goods sold is 60% of sales revenue), how much is ending inventory?

Answer:

$$\begin{aligned} \text{Beginning Inventory} + \text{Purchases} - \text{Ending Inventory} &= \text{Cost of Goods Sold} \\ \$70,000 + \$365,000 - \text{Ending Inventory} &= 60\% \text{ of } \$500,000 \\ \text{Ending Inventory} &= \$70,000 + \$365,000 - \$300,000 = \$135,000 \end{aligned}$$

The following Decision Guidelines summarize the situations that call for (a) a particular inventory system and (b) the motivation for using each costing method.

Analyzing Financial Statements

Using the Cost-of-Goods-Sold Model. While some may view this model as related to the periodic entry system, the **cost-of-goods-sold model** is used by all companies, regardless of their accounting system. The model is extremely powerful because it captures all the inventory information for an entire accounting period.

A1		
	A	B
1	Cost of goods sold (in thousands):	
2	Beginning inventory	\$XXXX
3	+ Purchases	XXXX
4	= Cost of goods available	XXXX
5	– Ending inventory	(XXXX)
6	= Cost of goods sold	\$XXXX
7		

While the model above shows the cost of goods sold on the last line, you can rework it to calculate any missing figure from the model.

Estimating Inventory by the Gross Profit Method. Often a business must *estimate* the value of its goods. The **gross profit method**, also known as the **gross margin method**, is widely used to estimate ending inventory. While this method uses the cost-of-goods-sold model, but for the gross-profit method, we rearrange the ending inventory and cost of goods sold as follows:

Beginning inventory	\$ XXXX
+ Purchases	XXXX
= Cost of goods available	XXXX
– Cost of goods sold.....	(XXXX)
= Ending inventory.....	\$ XXXX

Suppose a fire destroys some store's inventory as well as the computer system on which its records are kept, which prevents the manager from physically reconstructing the records of the inventory lost. To collect insurance, the store must estimate the cost of the ending inventory lost. Let's assume the store uses a *gross profit percentage* of 35%. We can estimate the store's cost of goods sold. Then subtract cost of goods sold from cost of goods available to estimate the amount of ending inventory. The calculations for the gross profit method are shown below, with new amounts assumed for the illustration.

You can also use the gross profit method to test the overall reasonableness of an ending inventory amount. This method also helps to detect large errors.

A1			
	A	B	C
1	Beginning inventory		\$ XXXX
2	Purchases		XXXX
3	Cost of goods available		XXXX
4	Estimated cost of goods sold:		
5	Net sales revenue	\$ XXXX	
6	Less estimated gross profit of 35%	XXXX	
7	Estimated cost of goods sold		XXXX
8	Estimated cost of ending inventory lost		\$ XXXX
9			



DECISION GUIDELINES

ACCOUNTING FOR INVENTORY

Suppose Franc Franc, a furniture store, stocks two basic categories of merchandise:

- furniture pieces, such as tables and chairs
- small items of low value, near the checkout stations, such as cupholders and coasters

Jacob Stiles, the store manager, is considering how accounting will affect the business. Let's examine several decisions Stiles must make to properly account for the store's inventory.

Decision	Guidelines	System or Method
Which inventory system to use?	• Expensive merchandise • Cannot control inventory by visual inspection	→ Perpetual system for the furniture
	• Can control inventory by visual inspection	→ Periodic system for the small, low-value items
Which costing method to use?	• Unique inventory items	→ Specific unit cost for designer furniture because they are unique
	• Most current cost of ending inventory • Maximizes reported income when costs are rising	→ FIFO
	• Most current measure of cost of goods sold and net income • Minimizes income tax when costs are rising	→ LIFO (not allowed under IFRS)
	• Middle-of-the-road approach for income tax and reported income	→ Average

End-of-Chapter

Summary Problem

Town & Country Gift Ideas began 20X6 with 60,000 units of inventory that cost \$36,000. During 20X6, Town & Country purchased merchandise on account for \$352,500 as follows:

Purchase 1	(100,000 units costing)	\$ 65,000
Purchase 2	(270,000 units costing)	175,500
Purchase 3	(160,000 units costing)	112,000

Cash payments on account totaled \$326,000 during the year.

Town & Country's sales during 20X6 consisted of 520,000 units of inventory for \$660,000, all on account. The company uses the FIFO inventory method.

Cash collections from customers were \$630,000. Operating expenses totaled \$240,500, of which Town & Country paid \$211,000 in cash. Town & Country credited Accrued Liabilities for the remainder. At December 31, Town & Country accrued income tax expense at the rate of 35% of income before tax.

Requirements

1. Make summary journal entries to record Town & Country's transactions for the year, assuming the company uses a perpetual inventory system.
2. Determine the FIFO cost of Town & Country's ending inventory at December 31, 20X6, in two ways:
 - a. Use a T-account.
 - b. Multiply the number of units on hand by the unit cost.
3. Show how Town & Country would compute cost of goods sold for 20X6. Follow the FIFO example on page 339.
4. Prepare Town & Country's Income Statement for 20X6. Show totals for the gross profit and income before tax.
5. Determine Town & Country's gross profit percentage, rate of inventory turnover, and net income as a percentage of sales for the year. In Town & Country's industry, a gross profit percentage of 40%, an inventory turnover of six times per year, and a net income percentage of 7% are considered excellent. How well does Town & Country compare to these industry averages?

Answers

Requirement 1

A1			
	A	B	C
1	Inventory (\$65,000 + \$175,500 + \$112,000)	352,500	
2	Accounts Payable		352,500
3			
4	Accounts Payable	326,000	
5	Cash		326,000
6			
7	Accounts Receivable	660,000	
8	Sales Revenue		660,000
9			
10	Cost of Goods Sold (see Requirement 3)	339,500	
11	Inventory		339,500
12			
13	Cash	630,000	
14	Accounts Receivable		630,000
15			
16	Operating Expenses	240,500	
17	Cash		211,000
18	Accrued Liabilities		29,500
19			
20	Income Tax Expense (see Requirement 4)	28,000	
21	Income Tax Payable		28,000
22			

Requirement 2

Inventory			
Beg. bal.	36,000		
Purchases	352,500	Cost of goods sold	339,500
End. bal.	49,000		

Number of units in ending inventory (60,000 + 100,000 + 270,000 + 160,000 – 520,000)		70,000
Unit cost of ending inventory at FIFO	× \$ 0.70	
(\$112,000 ÷ 160,000 from Purchase 3).....		
FIFO cost of ending inventory		<u>\$49,000</u>

Requirement 3

Cost of goods sold (520,000 units):	
60,000 units costing	\$ 36,000
100,000 units costing	65,000
270,000 units costing	175,500
90,000 units costing \$0.70 each*	63,000
Cost of goods sold.....	<u>\$339,500</u>

*From Purchase 3: \$112,000/160,000 units = \$0.70 per unit.

Requirement 4

A1		
	A	B
1	Town & Country Gift Ideas	Year Ended
2	Income Statement	Dec. 31, 20X6
3	Sales revenue	\$ 660,000
4	Cost of goods sold	339,500
5	Gross profit	320,500
6	Operating expenses	240,500
7	Income before tax	80,000
8	Income tax expense (35%)	28,000
9	Net income	<u>\$ 52,000</u>

Requirement 5

		Industry Average
Gross profit percentage:	$\$320,500 \div \$660,000 = 48.6\%$	40%
Inventory turnover:	$\frac{\$339,500}{(\$36,000 + \$49,000)/2} = 8 \text{ times}$	6 times
Net income as a percent of sales:	$\$52,000 \div \$660,000 = 7.9\%$	7%

Town & Country's statistics are better than the industry averages.

REVIEW | Inventory and Merchandising Operations

Quick Check (Answers are given at the end of the chapter.)

- Which statement is true?
 - The invoice is the purchaser's request for collection from the customer.
 - A service company purchases products from suppliers and then sells them.
 - Gross profit is the excess of sales revenue over cost of goods sold.
 - The Sales account is used to record only sales on account.
- Sales discounts should appear in the financial statements
 - as an addition to inventory.
 - as a deduction from sales.
 - among the current liabilities.
 - as an addition to sales.
 - as an operating expense.
- How is inventory classified in the financial statements?
 - As a liability
 - As a contra account to Cost of Goods Sold
 - As an expense
 - As an asset
 - As a revenue

Questions 4–6 use the following data of Dana, Inc.

	Units	Unit Cost	Total Cost	Units Sold
Beginning inventory	15	\$6	\$ 90	
Purchase on Apr 25	40	7	280	
Purchase on Nov 13	10	9	90	
Sales	40	?	?	

- Dana uses a FIFO inventory system. Cost of goods sold for the period is
 - \$300.
 - \$283.
 - \$265.
 - \$195.
- Dana's LIFO cost of ending inventory would be
 - \$195.
 - \$160.
 - \$177.
 - \$175.
- Dana's average cost of ending inventory is
 - \$195.
 - \$160.
 - \$177.
 - \$175.
- When applying the lower of cost or net realizable value, NRV means
 - selling price less discounts.
 - original cost plus profit margin.
 - selling price less cost to sell.
 - original cost less physical deterioration.
- During a period of rising prices, the inventory method that will yield the highest net income and asset value is
 - LIFO.
 - FIFO.
 - specific identification.
 - average cost.

9. Which statement is true?
- The inventory method that best matches current expense with current revenue is FIFO.
 - When prices are rising, the inventory method that results in the lowest ending inventory value is FIFO.
 - An error overstating ending inventory in 2010 will understate 2010 net income.
 - Application of the inventory valuation rule may result in a lower inventory value than the cost of inventory.
10. The ending inventory of Misty Harbor Co. is \$75,000. If beginning inventory was \$86,000 and goods available totaled \$127,000, the cost of goods sold is
- \$41,000.
 - \$52,000.
 - \$86,000.
 - \$138,000.
 - none of the above.
11. Lantern Company had cost of goods sold of \$148,000. The beginning and ending inventories were \$16,000 and \$28,000, respectively. Purchases for the period must have been
- \$136,000.
 - \$160,000.
 - \$164,000.
 - \$176,000.
 - \$192,000.

Use the following information for questions 12–14.

Highway Company had a \$28,000 beginning inventory and a \$35,000 ending inventory. Net sales were \$184,000; purchases, \$93,000; purchase returns and allowances, \$7,000; and freight-in, \$3,000.

12. Cost of goods sold for the period is
- \$96,000.
 - \$98,000.
 - \$82,000.
 - \$81,000.
 - none of the above.
13. What is Highway's gross profit percentage (rounded to the nearest percentage)?
- 50%
 - 55%
 - 45%
 - None of the above
14. What is Highway's rate of inventory turnover?
- 2.6 times
 - 5.3 times
 - 3.0 times
 - 2.3 times
15. Beginning inventory is \$120,000, purchases are \$270,000, and sales total \$460,000. The normal gross profit is 40%. Using the gross profit method, how much is ending inventory?
- \$206,000
 - \$132,000
 - \$114,000
 - \$88,000
 - None of the above
16. An overstatement of ending inventory in one period results in
- an understatement of the beginning inventory of the next period.
 - an understatement of net income of the next period.
 - an overstatement of net income of the next period.
 - no effect on net income of the next period.

Accounting Vocabulary

average-cost method (p. 344) Inventory costing method based on the average cost of inventory during the period. Average cost is determined by dividing the cost of goods available by the number of units available. Also called the *weighted-average method*.

consignment (p. 332) An inventory arrangement where the seller sells inventory that belongs to another party. The seller does not include consigned merchandise on hand in its Balance Sheet, because the seller does not own this inventory.

comparability (p. 345) A business must use the same accounting methods and procedures from period to period. Comparability is an enhancing qualitative characteristic of financial statements.

cost formulas (p. 338) Cost assumptions (e.g., LIFO, FIFO, or weighted-average) used in the measurement of inventory and cost of goods sold when goods are ordinarily interchangeable. Also called *inventory costing method*.

cost of goods sold (p. 330) Cost of the inventory the business has sold to customers. Also known as cost of sales.

cost-of-goods-sold model (p. 352) Formula that brings together all the inventory data for the entire accounting period: Beginning inventory + Purchases = Goods available. Then, Goods available – Ending inventory = Cost of goods sold.

cost of sales (p. 330). A synonym for *cost of goods sold*.

debit memorandum (p. 336) A document issued to the seller (vendor) when an item of inventory that is unwanted or damaged is returned. This document authorizes a reduction (debit) to accounts payable for the amount of the goods returned.

first-in, first-out (FIFO) cost (method) (p. 338) Inventory costing method by which the first costs into inventory are the first costs out to cost of goods sold. Ending inventory is based on the costs of the most recent purchases.

FOB (p. 332) Stands for *free on board* (or *freight on board*), a legal term that designates the point at which title passes for goods sold. FOB shipping point means that the buyer owns, and therefore is legally obligated to pay for, goods at the point of shipment, including transportation costs. In this case, the buyer owns the goods while they are in transit from the seller and must include their costs, including freight, in inventory at that point. FOB destination means that the seller pays the transportation costs, so the goods do not belong to the buyer until they reach the buyer's place of business.

gross margin (p. 332) Another name for *gross profit*.

gross margin percentage (p. 349) Another name for the *gross profit percentage*.

gross profit (p. 332) Sales revenue minus cost of goods sold. Also called *gross margin*.

gross profit method (p. 352) A way to estimate inventory based on a rearrangement of the cost-of-goods-sold model: Beginning inventory + Net purchases = Goods available – Cost of goods sold = Ending inventory. Also called the *gross margin method*.

gross profit percentage (p. 349) Gross profit divided by net sales revenue. Also called the *gross margin percentage*.

inventory (p. 330) The merchandise that a company sells to customers.

inventory costing method (p. 338) Cost assumptions (e.g. LIFO, FIFO, or weighted-average) used in the measurement of inventory and cost of goods sold when goods are ordinarily interchangeable. Also called *cost formulas*.

inventory resident period (p. 351) The average number of days to sell inventory to customers, calculated by dividing 365 by inventory turnover. Also called *days inventory on hand*.

inventory turnover (p. 350) The number of times inventories are sold during the year, approximated by taking total cost of goods sold divided by average inventory. Used in calculating *inventory resident period*.

last-in, first-out (LIFO) cost (method) (p. 338) Inventory costing method by which the last costs into inventory are the first costs out to cost of goods sold. This method leaves the oldest costs—those of beginning inventory and the earliest purchases of the period—in ending inventory.

net realizable value (NRV) (p. 346) The estimated selling price in the ordinary course of business less the estimated costs of completion and the estimated costs necessary to make the sale.

periodic inventory system (p. 334) An inventory system in which the business does not keep a continuous record of the inventory on hand. Instead, at the end of the period, the business makes a physical count of the inventory on hand and applies the appropriate unit costs to determine the cost of the ending inventory.

perpetual inventory system (p. 334) An inventory system in which the business keeps a continuous record for each inventory item to show the inventory on hand at all times.

purchase allowance (p. 336) A decrease in the cost of purchases because the seller has granted the buyer a subtraction (an allowance) from the amount owed.

purchase discount (p. 335) A decrease in the cost of purchases earned by making an early payment to the vendor.

purchase return (p. 336) A decrease in the cost of purchases because the buyer returned the goods to the seller.

specific identification (p. 338) When goods are ordinarily not interchangeable, cost of goods sold and on-hand are determined by specifically identifying which units have been sold and which units remain on-hand, respectively.

weighted-average cost method (p. 338) Another name for the *average-cost method*.

ASSESS YOUR PROGRESS

Short Exercises

S6-1. (Learning Objective 2: Accounting for inventory transactions) Journalize the following assumed transactions for The Debson Company. Show amounts in millions.

L0 2

- Cash purchases of inventory, \$3.6 million
- Sales on account, \$19.5 million
- Cost of goods sold (perpetual inventory system), \$4.8 million
- Collections on account, \$19.1 million

S6-2. (Learning Objective 2: Accounting for inventory transactions) Summer Kluxon, Inc., purchased inventory costing \$150,000 and sold 75% of the goods for \$180,000. All purchases and sales were on account. Kluxon later collected 40% of the accounts receivable.

L0 2

- Journalize these transactions for Kluxon, which uses the perpetual inventory system.
- For these transactions, show what Kluxon will report for inventory, revenues, and expenses on its financial statements. Report gross profit on the appropriate statement.

S6-3. (Learning Objective 3: Applying the average, FIFO, and LIFO methods) Coastal Sporting Goods started April with an inventory of nine sets of golf clubs that cost a total of \$1,620. During April, Coastal purchased 25 sets of clubs for \$4,750. At the end of the month, Coastal had eight sets of golf clubs on hand. The store manager must select an inventory costing method, and he asks you to tell him both cost of goods sold and ending inventory under these three accounting methods:

L0 3

- Average cost (round average unit cost to the nearest cent)
- FIFO
- LIFO

S6-4. (Learning Objective 2: Applying the average, FIFO, and LIFO methods) Jackson's Copy Center uses laser printers. Assume Jackson started the year with 92 containers of ink (average cost of \$9.20 each, FIFO cost of \$8.75 each, LIFO cost of \$8.30 each). During the year, Jackson purchased 680 containers of ink at \$10.30 and sold 580 units for \$22.50 each. Jackson paid operating expenses throughout the year, a total of \$3,570. Jackson is not subject to income tax.

L0 2

Prepare Jackson's Income Statement for the current year ended December 31 under the average, FIFO, and LIFO inventory costing methods. Include a complete statement heading.

S6-5. (Learning Objective 3: Computing income tax effects of the inventory costing methods) This exercise should be used in conjunction with Short Exercise 6-4. Jackson is a corporation subject to a 35% income tax. Compute Jackson's income tax expense under the average, FIFO, and LIFO inventory costing methods. Which method would you select to (a) maximize income before tax and (b) minimize income tax expense?

L0 3

S6-6. (Learning Objective 3: Computing income and income tax effects of LIFO)

L0 3

Macrodata.com uses the LIFO method to account for inventory. Macrodata is having an unusually good year, with net income well above expectations. The company's inventory costs are rising rapidly. What can Macrodata do immediately before the end of the year to decrease net income? Explain how this action decreases reported income, and also why Macrodata might want to decrease its net income.

■ writing assignment

S6-7. (Learning Objective 3: Applying the Net Realizable Value to inventory) It is December 31, end of the year, and the controller of Grass Corporation is applying the Net Realizable Value (NRV) to inventories. Before any year-end adjustments, Grass reports the following data:

L0 3

Cost of goods sold.....	\$470,000
Historical cost of ending inventory, as determined by a physical count	54,000

■ writing assignment

L0 3

Grass determines that the NRV of ending inventory is \$48,000. Show what Grass should report for ending inventory and for cost of goods sold. Identify the financial statement where each item appears.

S6-8. (Learning Objective 3: Managing income taxes under the LIFO method) Stone Saxophone Company is nearing the end of its worst year ever. With two weeks until year-end, it appears that net income for the year will have decreased by 25% from last year. Jason Stone, the president and principal shareholder, is distressed with the year's results. Stone asks you, the financial vice president, to come up with a way to increase the business's net income. Inventory quantities are a little higher than normal because sales have been slow during the last few months. Stone uses the LIFO inventory method, and inventory costs have risen dramatically during the latter part of the year.

Complete the memorandum to Jason Stone to explain how the company can increase its net income for the year. Explain your reasoning in detail. Stone is a man of integrity, so your plan must be completely ethical.

L0 3

S6-9. (Learning Objective 3: Identifying income, tax, and other effects of the inventory methods) This exercise tests your understanding of the four inventory methods. List the name of the inventory method that best fits the description. Assume that the cost of inventory is rising.

1. Matches the most current cost of goods sold against sales revenue.
2. Results in an old measure of the cost of ending inventory.
3. Writes inventory down when replacement cost drops below historical cost.
4. Enables a company to buy high-cost inventory at year-end and thereby decrease reported income and income tax.
5. Enables a company to keep reported income from dropping lower by liquidating older layers of inventory.
6. Is generally associated with saving income taxes.
7. Results in a cost of ending inventory that is close to the current cost of replacing the inventory.
8. Is used to account for automobiles, jewelry, and art objects.
9. Provides a middle-ground measure of ending inventory and cost of goods sold.
10. Maximizes reported income.

L0 4

S6-10. (Learning Objective 4: Using ratio data to evaluate operations) Mountain Company made sales of \$38,542 million during 20X6. Cost of goods sold for the year totaled \$16,543 million. At the end of 20X5, Mountain's inventory stood at \$1,461 million, and Mountain ended 20X6 with inventory of \$2,045 million.

Compute Mountain's gross profit percentage and rate of inventory turnover for 20X6.

L0 4

S6-11. (Learning Objective 4: Estimating ending inventory by the gross profit method) Meteor Technology began the year with inventory of \$254,000 and purchased \$1,580,000 of goods during the year. Sales for the year are \$4,200,000, and Meteor's gross profit percentage is 60% of sales. Compute Meteor's estimated cost of ending inventory by using the gross profit method.

L0 5

S6-12. (Learning Objective 5: Assessing the effect of an inventory error—one year only) XYZ, Inc., reported these figures for its fiscal year (amounts in millions):

Net sales.....	\$ 2,100
Cost of goods sold.....	1,150
Ending inventory.....	480

Suppose XYZ later learns that ending inventory was overstated by \$14 million. What are the correct amounts for (a) net sales, (b) ending inventory, (c) cost of goods sold, and (d) gross profit?

L0 5

S6-13. (Learning Objective 5: Assessing the effect of an inventory error on 2 years) Kinder's \$5.8 million cost of inventory at the end of last year was understated by \$1.9 million.

1. Was last year's reported gross profit of \$3.2 million overstated, understated, or correct? What was the correct amount of gross profit last year?
2. Is this year's gross profit of \$5.6 million overstated, understated, or correct? What is the correct amount of gross profit for the current year?

S6-14. (Learning Objectives 2, 4: Considering ethical implications of inventory actions)

L0 2 4

Determine whether each of the following actions in buying, selling, and accounting for inventories is ethical or unethical. Give your reason for each answer.

1. Wellness Pharmaceuticals purchased lots of inventory shortly before year-end to increase the LIFO cost of goods sold and decrease reported income for the year.
2. In applying the net realizable value to inventories, Rich Financial Industries recorded an excessively low market value for ending inventory. This allowed the company to pay less income tax for the year.
3. Geller Corporation deliberately overstated purchases to produce a high figure for cost of goods sold (low amount of net income). The real reason was to decrease the company's income tax payments to the government.
4. Ditto Sales Company deliberately overstated ending inventory in order to report higher profits (net income).
5. Blueberry, Inc., delayed the purchase of inventory until after December 31, 20X6, to keep 20X6's cost of goods sold from growing too large. The delay in purchasing inventory helped net income of 20X6 to reach the level of profit demanded by the company's investors.

Exercises MyLab Accounting

Select A and B exercises can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select exercises using MyLab Accounting.

Group A

E6-15A. (Learning Objectives 1, 2: Accounting for inventory transactions under FIFO costing) Accounting records for Richard Corporation yield the following data for the year ended December 31, 20X6:

L0 1 2

Inventory, December 31, 20X5	\$ 9,000
Purchases of inventory (on account).....	48,000
Sales of inventory—79% on account; 21% for cash (cost \$41,000).....	77,000
Inventory at FIFO, December 31, 20X6	16,000

Requirements

1. Journalize Richard's inventory transactions for the year under the perpetual system.
2. Report ending inventory, sales, cost of goods sold, and gross profit on the appropriate financial statement.

E6-16A. (Learning Objectives 2, 3: Analyzing inventory transactions under FIFO costing) Kurt's, Inc.'s inventory records for a particular development program show the following at December 31:

L0 2 3

■ spreadsheet

Dec 1	Beginning inventory	5 units @ \$160 = \$ 800
15	Purchase.....	4 units @ \$165 = \$ 660
26	Purchase.....	12 units @ \$180 = \$ 2,160

At December 31, nine of these programs are on hand. Journalize for Kurt's:

1. Total December purchases in one summary entry. All purchases were on credit.
2. Total December sales and cost of goods sold in two summary entries. The selling price was \$520 per unit, and all sales were on credit. Assume that Kurt's uses the FIFO inventory method and the sale took place on December 28.
3. Under FIFO, how much gross profit would Kurt's earn on these transactions? What is the FIFO cost of Kurt's ending inventory?

L0 2 3

E6-17A. (Learning Objective 2, 3: Determining ending inventory and cost of goods sold by four methods) Use the data for Kurt's in Exercise 6-16A to answer the following:

Requirements

1. Compute cost of goods sold and ending inventory, using each of the following methods:
 - a. Specific unit cost, with three \$160 units and six \$180 units still on hand at the end.
 - b. Average cost.
 - c. First-in, first-out.
 - d. Last-in, first-out.
2. Which method produces the highest cost of goods sold? Which method produces the lowest cost of goods sold? What causes the difference in cost of goods sold?

L0 2 3

E6-18A. (Learning Objective 2, 3: Computing the tax advantage of LIFO over FIFO) Use the data for Kurt's in Exercise 6-16A to illustrate Kurt's income tax advantage from using LIFO over FIFO. Sales revenue is \$6,800, operating expenses are \$1,200, and the income tax rate is 35%. How much in taxes would Kurt's save by using the LIFO method versus FIFO?

L0 2 3

E6-19A. (Learning Objective 2, 3: Determining ending inventory and cost of goods sold—FIFO vs. LIFO) MusicSheet.net specializes in sound equipment. Because each inventory item is expensive, MusicSheet uses a perpetual inventory system. Company records indicate the following data for a line of speakers:

Date	Item	Quantity	Unit Cost	Sale Price
Apr 1	Balance.....	18	\$34	
Apr 2	Purchase.....	6	65	
Apr 7	Sale	9		\$122
Apr 13	Sale	6		122

Requirements

1. Determine the amounts that MusicSheet should report for cost of goods sold and ending inventory in two ways:
 - a. FIFO
 - b. LIFO
2. MusicSheet uses the FIFO method. Prepare MusicSheet's Income Statement for the month ended April 30, reporting gross profit. Operating expenses totaled \$280, and the income tax rate was 32%.

L0 3

E6-20A. (Learning Objective 3: Measuring gross profit—FIFO vs. LIFO; Falling prices) Suppose a Waldorf store in Atlanta, Georgia, ended November 20X6 with 800,000 units of merchandise that cost an average of \$5 each. Suppose the store then sold 600,000 units for \$5.2 million during December. Further, assume the store made two large purchases during December as follows:

Dec 11	200,000 units @ \$4.00	= \$ 800,000
24	500,000 units @ \$3.00	= \$1,500,000

Requirements

1. At December 31, the store manager needs to know the store's gross profit under both FIFO and LIFO. Supply this information.
2. What caused the FIFO and LIFO gross profit figures to differ?
3. Assume that the store uses FIFO to value inventories, and that the store manager, whose bonus is based on profits, decides to change the unit cost on inventory to \$5 for all units. What impact will this have on gross profit and net income? Should this be allowed?

E6-21A. (Learning Objective 3: Applying the net realizable value to inventories) Lush Garden Supplies uses a perpetual inventory system. Lush Garden Supplies has these account balances at July 31, 20X6, prior to making the year-end adjustments:

L0 3

Inventory		Cost of Goods Sold		Sales Revenue	
Beg. bal.	11,500				
End. bal.	18,000	Bal.	76,000	Bal.	161,000

A year ago, the NRV cost of ending inventory was \$12,000, which exceeded cost of \$11,500. Lush Garden Supplies has determined that the NRV cost of the July 31, 20X6, ending inventory is \$14,800.

Requirement

1. Prepare Lush Garden Supplies' 20X6 Income Statement through gross profit to show how the company would apply the net realizable value to its inventories.

E6-22A. (Learning Objective 4: Using the cost-of-goods-sold model) Supply the missing Income Statement amounts for each of the following companies:

L0 4

Company	Net Sales	Beginning Inventory	Net Purchases	Ending Inventory	Cost of Goods Sold	Gross Profit
Chris	\$125,000	\$21,000	\$65,000	\$19,000	(a)	(b)
Ford	138,000	26,000	(c)	27,000	(d)	44,000
Arthur	(e)	(f)	65,000	32,000	60,000	38,000
Michaels	94,000	11,000	32,000	(g)	38,000	(h)

Requirement

1. Prepare the Income Statement for Chris Company, for the year ended December 31. Use the cost-of-goods-sold model to compute cost of goods sold. Chris's operating and other expenses, as adapted, for the year were \$42,000. Ignore income tax.

E6-23A. (Learning Objective 4: Measuring profitability) Refer to the data in Exercise 6-22A. Compute all ratio values to answer the following questions:

L0 4

- Which company has the highest, and which company has the lowest, gross profit percentage?
- Which company has the highest, and which the lowest, rate of inventory turnover? Based on your figures, which company appears to be the most profitable?

E6-24A. (Learning Objective 4: Computing gross profit percentage and inventory turnover) Thick & Thin, a partnership, had the following inventory data:

L0 4

	20X5	20X6
Ending inventory at:		
FIFO Cost	\$22,000	\$ 24,000
LIFO Cost	10,000	16,000
Cost of goods sold at:		
FIFO Cost		\$ 82,500
LIFO Cost		98,200
Sales revenue		148,000

Thick & Thin need to know the company's gross profit percentage and rate of inventory turnover for 20X6 under:

1. FIFO
2. LIFO

Which method makes the business look better on:

3. Gross profit percentage?
4. Inventory turnover?

L0 4

E6-25A. (Learning Objective 4: Budgeting inventory purchases) Toys Plus prepares budgets to help manage the company. Toys Plus is budgeting for the fiscal year ended January 31, 20X6. During the preceding year ended January 31, 20X5, sales totaled \$9,400 million and cost of goods sold was \$6,700 million. At January 31, 20X5, inventory stood at \$2,200 million. During the upcoming 20X6 year, suppose Toys Plus expects cost of goods sold to increase by 10%. The company budgets next year's ending inventory at \$2,500 million.

Requirement

1. One of the most important decisions a manager makes is how much inventory to buy. How much inventory should Toys Plus purchase during the upcoming year to reach its budgeted figures?

L0 4

■ spreadsheet

E6-26A. (Learning Objective 4: Estimating inventory by the gross profit method) J C Company began May with inventory of \$48,500. The business made net purchases of \$32,900 and had net sales of \$64,000 before a fire destroyed the company's inventory. For the past several years, J C's gross profit percentage has been 35%. Estimate the cost of the inventory destroyed by the fire. Identify another reason owners and managers use the gross profit method to estimate inventory.

L0 5

E6-27A. (Learning Objective 5: Correcting an inventory error) Mighty Sea Marine Supply reported the following comparative Income Statement for the years ended September 30, 20X6 and 20X5:

A1				
	A	B	C	D
				E
1	Mighty Sea Marine Supply	Year Ended September 30		
2	Income Statements	20X6	20X5	
3	Sales revenue		\$ 143,000	\$ 120,000
4	Cost of goods sold:			
5	Beginning inventory	\$ 14,500		\$ 9,000
6	Net purchases	74,000		67,000
7	Cost of goods available	88,500		76,000
8	Ending inventory	(19,000)		(14,500)
9	Cost of goods sold		69,500	61,500
10	Gross profit		73,500	58,500
11	Operating expenses		28,000	24,000
12	Net income		\$ 45,500	\$ 34,500
13				

Mighty Sea's president and shareholders are thrilled by the company's boost in sales and net income during 20X6. Then the accountants for the company discover that ending 20X5 inventory was understated by \$7,500. Prepare the corrected comparative Income Statement for the two-year period, complete with a heading for the statement. How well did Mighty Sea really perform in 20X6, as compared with 20X5?

Group B

E6-28B. (*Learning Objectives 1, 2: Accounting for inventory transactions under FIFO costing*) Accounting records for Rockefeller Corporation yield the following data for the year ended December 31, 20X6:

L0 1 2

Inventory, December 31, 20X5	€ 7,000
Purchases of inventory (on account).....	45,000
Sales of inventory—76% on account; 24% for cash (cost \$40,000).....	72,000
Inventory at FIFO, December 31, 20X6	19,000

Requirements

1. Journalize Rockefeller's inventory transactions for the year under the perpetual system.
2. Report ending inventory, sales, cost of goods sold, and gross profit on the appropriate financial statement.

E6-29B. (*Learning Objectives 2, 3: Analyzing inventory transactions under FIFO costing*) Riva's, Inc.'s inventory records for a particular development program show the following at May 31:

L0 2 3

May 1	Beginning inventory	8 units @ \$150 =	\$1,200
15	Purchase.....	5 units @ \$150 =	750
26	Purchase.....	13 units @ \$160 =	2,080

At May 31, 10 of these programs are on hand. Journalize for Riva's:

1. Total May purchases in one summary entry. All purchases were on credit.
2. Total May sales and cost of goods sold in two summary entries. The selling price was \$640 per unit and all sales were on credit. Assume that Riva's uses the FIFO inventory method and the sale took place on May 28.
3. Under FIFO, how much gross profit would Riva's earn on these transactions? What is the FIFO cost of Riva's, Inc.'s ending inventory?

E6-30B. (*Learning Objective 2, 3: Determining ending inventory and cost of goods sold by four methods*) Use the data for Riva's, Inc., in Exercise 6-29B to answer the following.

L0 2 3

■ spreadsheet

Requirements

1. Compute cost of goods sold and ending inventory using each of the following methods:
 - a. Specific unit cost, with four \$150 units and six \$160 units still on hand at the end
 - b. Average cost
 - c. FIFO
 - d. LIFO
2. Which method produces the highest cost of goods sold? Which method produces the lowest cost of goods sold? What causes the difference in cost of goods sold?

E6-31B. (*Learning Objectives 2, 3: Computing the tax advantage of LIFO over FIFO*) Use the data for Riva's, Inc., in Exercise 6-29B to illustrate Riva's income tax advantage from using LIFO over FIFO. Sales revenue is \$8,570, operating expenses are \$2,300, and the income tax rate is 38%. How much in taxes would Riva's save by using the LIFO method versus FIFO?

L0 2 3

E6-32B. (*Learning Objectives 2, 3: Determining ending inventory and cost of goods sold—FIFO vs. LIFO*) MusicNotes.net specializes in sound equipment. Because each inventory item

L0 2 3

is expensive, MusicNotes uses a perpetual inventory system. Company records indicate the following data for a line of speakers:

Date	Item	Quantity	Unit Cost	Sale Price
Apr 1	Balance.....	17	\$38	
Apr 2	Purchase.....	5	68	
Apr 7	Sale	9		\$106
Apr 13	Sale	5		92

Requirements

- Determine the amounts that MusicNotes should report for cost of goods sold and ending inventory two ways:
 - FIFO
 - LIFO
- MusicNotes uses the FIFO method. Prepare MusicNotes' Income Statement for the month ended April 30, reporting gross profit. Operating expenses totaled \$430, and the income tax rate was 30%.

L0 3

E6-33B. (Learning Objective 3: Measuring gross profit—FIFO vs. LIFO; Falling prices) Suppose a store in Milan ended September with 1,500,000 units of merchandise that cost an average of €9.00 each. Suppose the store then sold 1,300,000 units for €14.1 million during October. Further, assume the store made two large purchases during October as follows:

Oct 12	100,000 units @ \$8.00 = € 800,000
24	600,000 units @ \$7.00 = €4,200,000

Requirements

- At October 31, the store manager needs to know the store's gross profit under both FIFO and LIFO. Supply this information.
- What caused the FIFO and LIFO gross profit figures to differ?
- Assume that the store uses FIFO, and that the store manager, whose bonus is based on profits, decides to value all units in ending inventory at €9 per unit. What impact will this action have on gross profit and net income? Should this be allowed?

L0 3

E6-34B. (Learning Objective 3: Applying the net realizable value to inventories) Secret Garden Supplies uses a perpetual inventory system. Secret Garden Supplies has these account balances at May 31, 20X6, prior to making the year-end adjustments:

Inventory		Cost of Goods Sold		Sales Revenue	
Beg. bal.	12,500				
End. bal.	15,300	Bal.	63,000		Bal. 152,000

A year ago, the NRV cost of ending inventory was €14,600, which exceeded the cost of €12,500. Secret Garden Supplies has determined that the NRV cost of the May 31, 20X6, ending inventory is €13,400.

Requirement

- Prepare Secret Garden Supplies' 20X6 Income Statement through gross profit to show how the company would apply the net realizable value to its inventories.

L0 4

E6-35B. (Learning Objective 4: Using the cost-of-goods-sold model) Supply the missing amounts for each of the following companies:

Company	Net Sales	Beginning Inventory	Net Purchases	Ending Inventory	Cost of Goods Sold	Gross Profit
Frank	€121,000	€18,000	€64,000	€17,000	(a)	(b)
Hill	135,000	26,000	(c)	28,000	(d)	48,000
Fort	(e)	(f)	56,000	22,000	61,000	32,000
Orville	89,000	8,000	33,000	(g)	36,000	(h)

Requirement

1. Prepare the Income Statement for Frank Company, for the year ended December 31. Use the cost-of-goods-sold model to compute cost of goods sold. Frank's operating and other expenses for the year were €48,000. Ignore income tax.

E6-36B. (Learning Objective 3: Measuring profitability) Refer to the data in Exercise 6-35B. Compute all ratio values to answer the following questions:

L0 3

- Which company has the highest, and which company has the lowest, gross profit percentage?
 - Which company has the highest, and which the lowest, rate of inventory turnover?
- Based on your figures, which company appears to be the most profitable?

E6-37B. (Learning Objective 3: Computing gross profit percentage and inventory turnover) Dunk & Dive, a partnership, had these inventory data:

L0 3

	20X5	20X6
Ending inventory at:		
FIFO Cost	\$19,000	\$ 22,000
LIFO Cost	16,000	23,000
Cost of goods sold at:		
FIFO Cost		\$ 87,500
LIFO Cost		95,200
Sales revenue		145,000

Dunk & Dive need to know the company's gross profit percentage and rate of inventory turnover for 20X6 under

1. FIFO
2. LIFO

Which method makes the business look better on

3. Gross profit percentage?
4. Inventory turnover?

E6-38B. (Learning Objective 4: Budgeting inventory purchases) Toyland prepares budgets to help manage the company. Toyland is budgeting for the fiscal year ended January 31, 20X6. During the preceding year ended January 31, 20X5, sales totaled €9,500 million and cost of goods sold was €6,400 million. At January 31, 20X5, inventory stood at €1,900 million. During the upcoming 20X6 year, suppose Toyland expects cost of goods sold to increase by 10%. The company budgets next year's inventory at €2,200 million.

L0 4

Requirement

1. One of the most important decisions a manager makes is how much inventory to buy. How much inventory should Toyland purchase during the upcoming year to reach its budgeted figures?

■ spreadsheet

LO 4

E6-39B. (*Learning Objective 4: Estimating inventory by the gross profit method*) R T Company began June with inventory of €45,700. The business made net purchases of €39,100 and had net sales of €68,400 before a fire destroyed the company's inventory. For the past several years, R T's gross profit percentage has been 40%. Estimate the cost of the inventory destroyed by the fire. Identify another reason owners and managers use the gross profit method to estimate inventory.

LO 5

E6-40B. (*Learning Objective 5: Correcting an inventory error*) Friendly Harbor Marine Supply reported the following comparative Income Statement for the years ended September 30, 20X6 and 20X5:

A1				
	A	B	C	D
	Friendly Harbor Marine Supply Income Statements			
	12 Months Ended September 30			
		20X6	20X5	
1				
2				
3	Sales revenue		€ 139,000	€ 121,000
4	Cost of goods sold:			
5	Beginning inventory	€ 13,000		€ 12,000
6	Net purchases	74,000		69,000
7	Cost of goods available	87,000		81,000
8	Ending inventory	(18,500)		(13,000)
9	Cost of goods sold		68,500	68,000
10	Gross profit		70,500	53,000
11	Operating expenses		26,000	19,000
12	Net income		€ 44,500	€ 34,000
13				

Friendly Harbor's president and shareholders are thrilled by the company's boost in sales and net income during 20X6. Then the accountants for the company discover that ending 20X5 inventory was understated by €6,800. Prepare the corrected comparative Income Statement for the two-year period, complete with a heading for the statement. How well did Friendly Harbor really perform in 20X6, as compared with 20X5?

Challenge Exercises

LO 1 3

E6-41. (*Learning Objectives 1, 3: Making inventory policy decisions*) For each of the following situations, identify the inventory method that you would use, or, given the use of a particular method, state the strategy that you would follow to accomplish your goal:

- Company management, like that of IBM and Esplanade Imports, prefers a middle-of-the-road inventory policy that avoids extremes.
- Inventory costs are *decreasing*, and your company's board of directors wants to minimize income taxes.
- Inventory costs are *increasing*, and the company prefers to report high income.
- Inventory costs have been stable for several years, and you expect costs to remain stable for the indefinite future. (Give the reason for your choice of method.)
- Inventory costs are increasing. Your company uses LIFO and is having an unexpectedly good year. It is near year-end, and you need to keep net income from increasing too much in order to save on income tax.
- Suppliers of your inventory are threatening a labor strike, and it may be difficult for your company to obtain inventory. This situation could increase your income taxes.

LO 3

E6-42. (*Learning Objective 3: Evaluating a company's profitability*) L Mart, Inc., declared bankruptcy. Let's see why. L Mart reported these figures:

A1					
	A	B	C	D	E
1	L Mart, Inc.	12 Months Ended Dec. 31			
2	Statement of Income				
3	(In millions of \$)	20X6	20X5	20X4	20X3
4	Sales	\$ 36.2	\$ 35.8	\$ 34.3	
5	Cost of sales	28.6	27.7	26.8	
6	Selling expenses	7.8	6.9	6.0	
7	Other expenses	0.2	1.0	0.8	
8	Net income (net loss)	\$ (0.4)	\$ 0.2	\$ 0.7	
9	Additional data:				
10	Ending inventory	\$ 8.9	\$ 7.3	\$ 7.8	\$ 6.4

Requirement

1. Evaluate the trend of L Mart's results of operations during 20X4 through 20X6. Consider the trends of sales, gross profit, and net income. Track the gross profit percentage (to three decimal places) and the rate of inventory turnover (to one decimal place) in each year. Also discuss the role that selling expenses must have played in L Mart's difficulties.

Quiz

Test your understanding of accounting for inventory by answering the following questions. Select the best choice from among the possible answers given.

The next two questions use the following facts. Oceanview Software began January with \$3,500 of merchandise inventory. During January, Oceanview made the following entries for its inventory transactions:

A1				
	A	B	C	D
1	Inventory		6,300	
2	Accounts Payable			6,300
3				
4	Accounts Receivable		7,600	
5	Sales Revenue			7,600
6				
7	Cost of Goods Sold		5,400	
8	Inventory			5,400
9				

Q6-43. How much was Oceanview's inventory at the end of January?

- a. \$5,400
- b. Zero
- c. \$4,400
- d. \$2,200

Q6-44. What was Oceanview's gross profit for January?

- a. Zero
- b. \$7,600
- c. \$5,400
- d. \$2,200

Q6-45. When does the cost of inventory become an expense?

- a. When inventory is delivered to a customer.
- b. When inventory is purchased from the supplier.
- c. When cash is collected from the customer.
- d. When payment is made to the supplier.

The next two questions use the following facts. Marble Frame Shop wants to know the effect of different inventory costing methods on its financial statements. Inventory and purchases data for April follow:

		Units	Unit Cost	Total Cost
Apr 1	Beginning inventory	2,600	\$15.00	\$39,000
4	Purchase	1,300	15.20	19,760
9	Sale	(1,500)		

Q6-46. If Marble Frame uses the FIFO method, the cost of the ending inventory will be

- a. \$39,000.
- b. \$36,160.
- c. \$36,260.
- d. \$36,000.

Q6-47. If Marble Frame uses the LIFO method, cost of goods sold will be

- a. \$22,600.
- b. \$22,760.
- c. \$22,500.
- d. \$19,760.

Q6-48. In a period of rising prices,

- a. Gross profit under FIFO will be higher than under LIFO.
- b. LIFO inventory will be greater than FIFO inventory.
- c. Net income under LIFO will be higher than under FIFO.
- d. Cost of goods sold under LIFO will be less than under FIFO.

Q6-49. The Income Statement for Feel Good Health Foods shows gross profit of \$154,000, operating expenses of \$128,000, and cost of goods sold of \$213,000. What is the amount of net sales revenue?

- a. \$367,000
- b. \$341,000
- c. \$495,000
- d. \$341,000

Q6-50. When the inventory cost is lower than NRV, the inventory should be reported at

- a. market price of inventory.
- b. replacement of inventory.
- c. selling price of inventory less cost to sell.
- d. acquisition cost of inventory.

Q6-51. The sum of (a) ending inventory and (b) cost of goods sold is

- a. goods available.
- b. beginning inventory.
- c. gross profit.
- d. net purchases.

Q6-52. The following data come from the inventory records of Draper Company:

Net sales revenue.....	\$632,000
Beginning inventory	68,000
Ending inventory.....	45,000
Net purchases.....	480,000

Based on these facts, the gross profit for Draper Company is

- a. \$152,000.
- b. \$175,000.
- c. \$129,000.
- d. Some other amount.

Q6-53. Ellen Braun Cosmetics ended the month of May with inventory of \$26,000. Ellen Braun expects to end June with inventory of \$14,000 after cost of goods sold of \$103,000. How much inventory must Ellen Braun purchase during June in order to accomplish these results?

- a. \$89,000
- b. \$91,000
- c. \$115,000
- d. Cannot be determined from the data given

Q6-54. Two financial ratios that clearly distinguish a discount chain such as Kmart from a high-end retailer such as Saks Fifth Avenue are the gross profit percentage and the rate of inventory turnover. Which set of relationships is most likely for Saks Fifth Avenue?

Gross profit percentage	Inventory turnover
a. Low	Low
b. Low	High
c. High	Low
d. High	High

Q6-55. Sales are \$540,000 and cost of goods sold is \$330,000. Beginning and ending inventories are \$29,000 and \$34,000, respectively. How many times did the company turn its inventory over during this period?

- | | |
|---------------|---------------|
| a. 17.1 times | c. 7.2 times |
| b. 6.7 times | d. 10.5 times |

Q6-56. Kruger, Inc., reported the following data:

Freight in.....	\$ 24,000	Sales returns.....	\$ 7,000
Purchases	215,000	Purchase returns.....	6,200
Beginning inventory	57,000	Sales revenue.....	460,000
Purchase discounts	4,800	Ending inventory.....	48,000

Kruger's gross profit percentage is

- | | |
|----------|----------|
| a. 46.1. | c. 47.7. |
| b. 51.7. | d. 50.0. |

Q6-57. Shailene Wood Company had the following beginning inventory, net purchases, net sales, and gross profit percentage for the first quarter of 20X6:

Beginning inventory, \$50,000	Net purchases, \$74,000
Net sales revenue, \$96,000	Gross profit rate, 50%

By the gross profit method, the ending inventory should be

- | | |
|--------------|--------------|
| a. \$74,000. | c. \$81,000. |
| b. \$76,000. | d. \$79,000. |

Q6-58. An error understated Regan Corporation's December 31, 20X6, ending inventory by \$42,000. What effect will this error have on total assets and net income for 20X6?

Assets	Net income
a. Understate	Understate
b. No effect	Overstate
c. Understate	No effect
d. No effect	No effect

Q6-59. An error understated Regan Corporation's December 31, 20X6, ending inventory by \$42,000. What effect will this error have on net income for 20X7?

- | | |
|---------------|--------------|
| a. Understate | c. No effect |
| b. Overstate | |

Problems MyLab Accounting

Select A and B problems can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select problems using MyLab Accounting

Group A

L0 1 2

P6-60A. (*Learning Objectives 1, 2: Accounting for inventory in a perpetual system using average costing method*) Nice Buy purchases inventory in crates of merchandise; each crate of inventory is a unit. The fiscal year of Nice Buy ends each February 28. Assume you are dealing with a single Nice Buy store in Taipei, Taiwan. The Taiwan store began 20X6 with an inventory of 23,000 units that cost a total of \$1,150,000. During the year, the store purchased merchandise on account as follows:

April (31,000 units at \$51).....	\$1,581,000
August (51,000 units at \$55).....	2,805,000
November (61,000 units at \$61)	3,721,000
Total purchases	<u>\$8,107,000</u>

Cash payments on account totaled \$7,907,000. During fiscal year 20X6, the store sold 150,000 units of merchandise for \$14,700,000, of which \$4,800,000 was for cash and the balance was on account. Nice Buy uses the average cost method for inventories. Operating expenses for the year were \$3,550,000. Nice Buy paid 70% in cash and accrued the rest as accrued liabilities. The store accrued income tax at the rate of 30%.

Requirements

1. Make summary journal entries to record the store's transactions for the year ended February 28, 20X6. Nice Buy uses a perpetual inventory system.
2. Prepare a T-account to show the activity in the Inventory Account.
3. Prepare the store's Income Statement for the year ended February 28, 20X6. Show totals for gross profit, income before tax, and net income.

L0 1 2

P6-61A. (*Learning Objective 1, 2: Measuring cost of goods sold and ending inventory—perpetual system*) Assume a Jaguar Sports outlet store began October 20X6 with 52 pairs of running shoes that cost the store \$35 each. The sale price of these shoes was \$70. During October, the store completed these inventory transactions:

			Units	Unit Cost	Unit Sales Price
Oct 3	Sale	13	\$35	\$70	
8	Purchase	84	36		
11	Sale	39	35	70	
19	Sale	5	36	71	
24	Sale	40	36	71	
30	Purchase	24	37		

Requirements

1. The preceding data are taken from the store's perpetual inventory records. Which cost method does the store use? Explain how you arrived at your answer.
2. Determine the store's cost of goods sold for October. Also compute gross profit for October.
3. What is the cost of the store's October 31 inventory of running shoes?

L0 2 3

P6-62A. (*Learning Objectives 2, 3: Computing inventory by three methods—perpetual system*) Fatigues Surplus began October with 78 tents that cost \$19 each. During the month, Fatigues Surplus made the following purchases at cost:

Oct 4	99 tents @ \$20 = \$1,980
19	155 tents @ \$21 = 3,410
25	48 tents @ \$23 = 1,104

Fatigues Surplus sold 334 tents, and at October 31 the ending inventory consisted of 46 tents. The sale price of each tent was \$52.

Requirements

1. Determine the cost of goods sold and ending inventory amounts for October under the average cost, FIFO cost, and LIFO cost. Round average cost per unit to four decimal places, and round all other amounts to the nearest dollar.
2. Explain why cost of goods sold is highest under LIFO. Be specific.
3. Prepare Fatigues Surplus' Income Statement for October. Report gross profit. Operating expenses totaled \$5,400. Fatigues Surplus uses average costing for inventory. The income tax rate is 40%.

P6-63A. (Learning Objective 2, 3: Applying the different inventory costing methods—perpetual system) The records of Byron Aviation include the following accounts for inventory of aviation fuel at December 31 of the current year:

L0 2 3

■ writing assignment

Inventory					
Jan 1	Balance	780 units @ \$7.80	\$ 6,084		
Mar 6	Purchase	310 units @ \$7.90	2,449		
Jun 22	Purchase	8,360 units @ \$8.10	67,716		
Oct 4	Purchase	540 units @ \$9.20	4,968		
Sales Revenue					
				Dec 31	9,010 units \$132,447

Requirements

1. Prepare a partial Income Statement through gross profit under the average, FIFO, and LIFO methods. Round average cost per unit to four decimal places and all other amounts to the nearest dollar.
2. Which inventory method would you use to minimize income tax? Explain why this method causes income tax to be the lowest.

P6-64A. (Learning Objective 3: Applying the net realizable value to inventories—perpetual system) Everything Trade Mart has recently had lackluster sales. The rate of inventory turnover has dropped, and the merchandise is gathering dust. It is now December 31, 20X6, and the current NRV cost of Everything's ending inventory is \$78,000 below what Everything actually paid for the goods, which was \$210,000. Before any adjustments at the end of the period, the Cost of Goods Sold account has a balance of \$750,000.

L0 3

■ writing assignment

- a. What accounting action should Everything take in this situation?
- b. Give any journal entry required.
- c. At what amount should Everything report Inventory on the Balance Sheet?
- d. At what amount should the company report Cost of Goods Sold on the Income Statement?
- e. Discuss the accounting principle or concept that is most relevant to this situation.

LO 4

P6-65A. (*Learning Objective 4: Using gross profit percentage and inventory turnover to evaluate two companies*) Sprinkle Top and Coffee Shop are both specialty food chains. The two companies reported these figures, in millions:

A1	A	B	C
1	Sprinkle Top, Inc. Income Statement	12 Months Ended Dec. 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Revenues:		
4	Net sales	\$ 554	\$ 727
5			
6	Costs and Expenses:		
7	Cost of goods sold	487	598
8	Selling, general, and administrative expenses	63	59
9			

A1	A	B	C
1	Sprinkle Top, Inc. Balance Sheet	December 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Assets		
4	Current assets:		
5	Cash and cash equivalents	\$ 15	\$ 29
6	Receivables	31	42
7	Inventories	29	38
8			

A1	A	B	C
1	Coffee Shop Corporation Income Statement	12 Months Ended Dec. 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Net sales	\$ 7,720	\$ 6,310
4	Cost of goods sold	3,170	2,614
5	Selling, general, and administrative expenses	2,930	2,350
6			

A1	A	B	C
1	Coffee Shop Corporation Balance Sheet	December 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Assets		
4	Current assets:		
5	Cash and temporary investments	\$ 318	\$ 173
6	Receivables, net	232	185
7	Inventories	629	547
8			

Requirements

1. Compute the gross profit percentage and the rate of inventory turnover for Sprinkle Top and Coffee Shop for 20X6.
2. Based on these statistics, which company looks more profitable? Why? What other expense category should we consider in evaluating these two companies?

P6-66A. (Learning Objectives 1, 4: Estimating inventory by the gross profit method; preparing the Income Statement) Assume Theon Company, a copy center, lost some inventory in a fire. To file an insurance claim, Theon Company must estimate its inventory by the gross profit method. Assume that for the past two years Theon Company's gross profit has averaged 40% of net sales. Suppose the Theon Company's inventory records reveal the following data:

L0 1 4

■ spreadsheet

Inventory, October 1	\$ 57,200
Transactions during October:	
Purchases	490,400
Purchase discounts	11,100
Purchase returns.....	70,800
Sales	668,000
Sales returns	11,500

Requirements

1. Estimate the cost of the lost inventory, using the gross profit method.
2. Prepare the October Income Statement for this product through gross profit. Show the detailed computations of cost of goods sold in a separate schedule.

P6-67A. (Learning Objective 3: Determining the amount of inventory to purchase)

Grammy's Convenience Store's Income Statement and Balance Sheet reported the following:

L0 3

A1	A	B
1	Grammy's Convenience Stores Income Statement Year Ended December 31, 20X5	
2	Sales	\$ 957,000
3	Cost of sales	720,000
4	Gross profit	237,000
5	Operating expenses	114,000
6	Net income	\$ 123,000
7		

A1	A	B	C	D
1	Grammy's Convenience Stores Balance Sheet December 31, 20X5			
2	Assets		Liabilities	
3	Cash	\$ 44,000	Accounts payable	\$ 31,000
4	Inventories	68,000	Note payable	187,000
5	Land and buildings, net	273,000	Total liabilities	218,000
6			Owner, capital	167,000
7	Total assets	\$ 385,000	Total liabilities and capital	\$ 385,000
8				

The business is organized as a proprietorship, so it pays no corporate income tax. The owner is budgeting for 20X6. He expects sales and cost of goods sold to increase by 8%. To meet customer demand, ending inventory will need to be \$78,000 at December 31, 20X6. The owner hopes to earn a net income of \$158,000 next year.

Requirements

1. One of the most important decisions a manager makes is the amount of inventory to purchase. Show how to determine the amount of inventory to purchase in 20X6.
2. Prepare the store's budgeted Income Statement for 20X6 to reach the target net income of \$158,000. To reach this goal, operating expenses must decrease by \$16,040.

L0 5

P6-68A. (*Learning Objective 5: Correcting inventory errors over a three-year period*) The accounting records of R.B. Video Sales show the data below (in millions). The shareholders are very happy with R.B.'s steady increase in net income.

Auditors discovered that the ending inventory for 20X4 was understated by \$3 million and that the ending inventory for 20X5 was also understated by \$3 million. The ending inventory at December 31, 20X6, was correct.

	20X6	20X5	20X4
Net sales revenue.....	\$43	\$40	\$37
Cost of goods sold:			
Beginning inventory	\$ 6	\$ 5	\$ 4
Net purchases	29	27	25
Cost of goods available	35	32	29
Less ending inventory.....	(7)	(6)	(5)
Cost of goods sold	28	26	24
Gross profit.....	15	14	13
Operating expenses	7	7	7
Net income.....	<u>\$ 8</u>	<u>\$ 7</u>	<u>\$ 6</u>

Requirements

1. Show corrected Income Statements for each of the three years.
2. How much did these assumed corrections add to or take away from R.B.'s total net income over the three-year period? How did the corrections affect the trend of net income?
3. Will R.B.'s shareholders still be happy with the company's trend of net income? Give the reason for your answer.

Group B

L0 1 2

P6-69B. (*Learning Objectives 1, 2: Accounting for inventory in a perpetual system using average costing method*) Best Guy purchases inventory in crates of merchandise; each crate of inventory is a unit. The fiscal year of Best Guy ends each February 28. Assume you are dealing with a single Best Guy store in Paris, France. The Paris store began 20X6 with an inventory of 18,000 units that cost a total of €990,000. During the year, the store purchased merchandise on account as follows:

April (33,000 units at €60).....	€1,980,000
August (53,000 units at €64).....	3,392,000
November (63,000 units at €70)	4,410,000
Total purchases.....	<u>€9,782,000</u>

Cash payments on account totaled €9,492,000. During fiscal 20X6, the store sold 158,000 units of merchandise for €15,484,000, of which €5,400,000 was for cash and the balance was on account. Best Guy uses the average cost method for inventories. Operating expenses for the year were €2,860,000. Best Guy paid 70% in cash and accrued the rest as accrued liabilities. The store accrued income tax at the rate of 35%.

Requirements

1. Make summary journal entries to record the store's transactions for the year ended February 28, 20X6. Best Guy uses a perpetual inventory system.
2. Prepare a T-account to show the activity in the Inventory Account.
3. Prepare the store's Income Statement for the year ended February 28, 20X6. Show totals for gross profit, income before tax, and net income.

P6-70B. (*Learning Objective 1, 2: Measuring cost of goods sold and ending inventory—perpetual system*) Assume a Championship Sports outlet store began March with 48 pairs of running shoes that cost the store \$38 each. The sale price of these shoes was \$66. During March the store completed these inventory transactions:

L0 1 2

		Units	Unit Cost	Unit Sale Price
Mar 3	Sale	18	\$38	\$66
8	Purchase.....	77	39	
11	Sale	30	38	66
19	Sale	12	39	69
24	Sale	37	39	69
30	Purchase.....	17	40	

Requirements

1. The preceding data are taken from the store's perpetual inventory records. Which cost method does the store use? Explain how you arrived at your answer.
2. Determine the store's cost of goods sold for March. Also compute gross profit for March.
3. What is the cost of the store's March 31 inventory of running shoes?

P6-71B. (*Learning Objectives 2, 3: Computing inventory by three methods—perpetual system*) SWAT Team Surplus began July with 69 tents that cost €24 each. During the month, SWAT Team Surplus made the following purchases at cost:

L0 2 3

Jul 4	108 tents @ €26 = €2,808
19	153 tents @ €28 = 4,284
25	38 tents @ €29 = 1,102

SWAT Team Surplus sold 316 tents, and at July 31 the ending inventory consists of 52 tents. The sale price of each tent was €55.

Requirements

1. Determine the cost of goods sold and ending inventory amounts for July under the average cost, FIFO cost, and LIFO cost. Round average cost per unit four decimal places, and round all other amounts to the nearest dollar.
2. Explain why cost of goods sold is highest under LIFO. Be specific.
3. Prepare a SWAT Team Surplus Income Statement for July. Report gross profit. Operating expenses totaled €3,500. SWAT Team Surplus uses average costing for inventory. The income tax rate is 32%.

P6-72B. (*Learning Objectives 2, 3: Applying the different inventory costing methods—perpetual system*) The records of Bryan Aviation include the following accounts for inventory of aviation fuel at December 31 of the current year:

L0 2 3

■ **writing assignment**

Inventory				
Jan 1	Balance	740 units @ €7.50	€ 5,550	
Mar 6	Purchase	320 units @ €7.80	2,496	
Jun 22	Purchase	8,380 units @ €8.10	67,878	
Oct 4	Purchase	530 units @ €9.10	4,823	
Sales Revenue				
		Dec 31	9,030 units	€128,226

Requirements

1. Prepare a partial Income Statement through gross profit under the average, FIFO, and LIFO methods. Round average cost per unit to four decimal places and all other amounts to the nearest whole dollar.
2. Which inventory method would you use to minimize income tax? Explain why this method causes income tax to be the lowest.

L0 3

■ writing assignment

P6-73B. (Learning Objective 3: Applying the net realizable value to inventories—perpetual system) Ariel Trade Mart has recently had lackluster sales. The rate of inventory turnover has dropped, and the merchandise is gathering dust. It is now December 31, 20X6, and the current NRV cost of Ariel's ending inventory is €72,000 below what Ariel actually paid for the goods, which was €270,000. Before any adjustments at the end of the period, the Cost of Goods Sold account has a balance of €830,000.

- a. What accounting action should Ariel take in this situation?
- b. Give any journal entry required.
- c. At what amount should Ariel report Inventory on the Balance Sheet?
- d. At what amount should the company report Cost of Goods Sold on the Income Statement?
- e. Discuss the accounting principle or concept that is most relevant to this situation.

L0 4

P6-74B. (Learning Objective 4: Using gross profit percentage and inventory turnover to evaluate two companies) Pastry People and Coffee Grind are both specialty food chains. The two companies reported these figures, in millions:

A1			
	A	B	C
1	Pastry People, Inc. Income Statement	12 Months Ended Dec. 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Revenues:		
4	Net sales	\$ 558	\$ 711
5			
6	Costs and Expenses:		
7	Cost of goods sold	486	595
8	Selling, general, and administrative expenses	62	56
9			

A1			
	A	B	C
1	Pastry People, Inc. Balance Sheet	December 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Assets		
4	Current assets:		
5	Cash and cash equivalents	\$ 18	\$ 26
6	Receivables	23	38
7	Inventories	18	31
8			

A1			
	A	B	C
1	Coffee Grind Corporation Income Statement	12 Months Ended Dec. 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Net sales	\$ 7,270	\$ 6,460
4	Cost of goods sold	3,290	2,702
5	Selling, general, and administrative expenses	2,965	2,380
6			

A1			
	A	B	C
1	Coffee Grind Corporation Balance Sheet	December 31	
2	(Adapted, in millions of \$)	20X6	20X5
3	Assets		
4	Current assets:		
5	Cash and temporary investments	\$ 311	\$ 178
6	Receivables, net	222	196
7	Inventories	632	548
8			

Requirements

1. Compute the gross profit percentage and the rate of inventory turnover for Pastry People and Coffee Grind for 20X6.
2. Based on these statistics, which company looks more profitable? Why? What other expense category should we consider in evaluating these two companies?

P6-75B. (*Learning Objectives 1, 4: Estimating inventory by the gross profit method; preparing the Income Statement*) Assume Joey Company, a sporting goods store, lost some inventory in a fire. To file an insurance claim, Joey Company must estimate its ending inventory by the gross profit method. Assume that for the past two years, Joey Company's gross profit has averaged 45% of net sales. Suppose Joey Company's inventory records reveal the following data:

L0 1 4

■ spreadsheet

Inventory, January 1	€ 57,700
Transactions during January:	
Purchases	490,800
Purchase discounts	12,200
Purchase returns.....	70,200
Sales	665,000
Sales returns	16,500

Requirements

1. Estimate the cost of the lost inventory, using the gross profit method.
2. Prepare the January Income Statement for this product through gross profit. Show the detailed computation of cost of goods sold in a separate schedule.

L0 3

P6-76B. (*Learning Objective 3: Determining the amount of inventory to purchase*) Chris' Convenience Store's Income Statement and Balance Sheet reported the following. The business is organized as a proprietorship, so it pays no corporate income tax. The owner is budgeting for 20X6. He expects sales and cost of goods sold to increase by 10%. To meet customer demand, ending inventory will need to be €76,000 at December 31, 20X6. The owner hopes to earn a net income of €161,000 next year.

A1	
A	B
Chris' Convenience Stores Income Statement Year Ended December 31, 20X5	
1	
2 Sales	€ 964,000
3 Cost of sales	722,000
4 Gross profit	242,000
5 Operating expenses	110,000
6 Net income	€ 132,000
7	

A1			
A	B	C	D
Chris' Convenience Stores Balance Sheet December 31, 20X5			
1			
2 Assets		Liabilities	
3 Cash	€ 35,000	Accounts payable	€ 28,000
4 Inventories	65,000	Note payable	193,000
5 Land and buildings, net	268,000	Total liabilities	221,000
6		Owner, capital	147,000
7 Total assets	€ 368,000	Total liabilities and capital	€ 368,000
8			

Requirements

1. One of the most important decisions a manager makes is the amount of inventory to purchase. Show how to determine the amount of inventory to purchase in 20X6.
2. Prepare the store's budgeted Income Statement for 20X6 to reach the target net income of €161,000. To reach this goal, operating expenses must decrease by €4,800.

L0 5

P6-77B. (*Learning Objective 5: Correcting inventory errors over a three-year period*) The accounting records of Waterville Video Sales show the data below (in millions). The shareholders are very happy with Waterville's steady increase in net income.

	20X6	20X5	20X4
Net sales revenue.....	€38	€35	€32
Cost of goods sold:			
Beginning inventory	€ 9	€ 8	€ 7
Net purchases	<u>32</u>	<u>30</u>	<u>28</u>
Cost of goods available	41	38	35
Less ending inventory.....	<u>(10)</u>	<u>(9)</u>	<u>(8)</u>
Cost of goods sold	<u>31</u>	<u>29</u>	<u>27</u>
Gross profit	<u>7</u>	<u>6</u>	<u>5</u>
Operating expenses	<u>3</u>	<u>3</u>	<u>3</u>
Net income.....	<u>\$ 4</u>	<u>\$ 3</u>	<u>\$ 2</u>

Auditors discovered that the ending inventory for 20X4 was understated by €2 million and that the ending inventory for 20X5 was also understated by €2 million. The ending inventory at December 31, 20X6, was correct.

Requirements

1. Show corrected Income Statements for each of the three years.
2. How much did these assumed corrections add to or take away from Waterville's total net income over the three-year period? How did the corrections affect the trend of net income?
3. Will Waterville's shareholders still be happy with the company's trend of net income? Give the reason for your answer.

APPLY YOUR KNOWLEDGE

Decision Cases

Case 1. (*Learning Objectives 1, 2: Assessing the impact of a year-end purchase of inventory*) Duracraft Corporation is nearing the end of its first year of operations. Duracraft made inventory purchases of \$745,000 during the year, as follows:

January	1,000 units @	\$100.00 =	\$100,000
July	4,000	121.25	485,000
November	<u>1,000</u>	160.00	<u>160,000</u>
Totals	6,000		\$745,000

Sales for the year are 5,000 units for \$1,200,000 of revenue. Expenses other than cost of goods sold and income taxes total \$200,000. The president of the company is undecided about whether to adopt the FIFO method or the LIFO method for inventories. The income tax rate is 40%.

Requirements

1. To aid company decision making, prepare Income Statements under FIFO and under LIFO.
2. Compare the net income under FIFO with net income under LIFO. Which method produces the higher net income? What causes this difference? Be specific.

Case 2. (*Learning Objective 3: Assessing the impact of the inventory costing method on the financial statements*) The inventory costing method a company chooses can affect the financial statements and thus the decisions of the people who use those statements.

L0 1 2

■ writing assignment

L0 3

■ writing assignment

Requirements

1. Company A uses the LIFO inventory method and discloses its use of the LIFO method in notes to the financial statements. Company B uses the FIFO method to account for its inventory. Company B does *not* disclose which inventory method it uses. Company B reports a higher net income than Company A. In which company would you prefer to invest? Give your reason.
2. Conservatism is an accepted accounting concept. Would you want management to be conservative in accounting for inventory if you were a shareholder or a creditor of a company? Give your reason.



Ethical Issue

During 20X6, Vanguard, Inc., changed to the LIFO method of accounting for inventory. Suppose that during 20X7, Vanguard changes back to the FIFO method and the following year Vanguard switches back to LIFO again.

Requirements

1. What would you think of a company's ethics if it changed accounting methods every year?
2. What accounting principle would changing methods every year violate?
3. Who can be harmed when a company changes its accounting methods too often? How?

Focus on Financials: | Nestlé



This case spans all 12 chapters and is based on the consolidated financial statements of Nestlé. As you work with Nestlé throughout this course, you will develop the confidence and ability to use the financial statements of other companies as well.

Refer to Nestlé financial statements in Appendix A. Alternatively, you may choose to obtain the full annual report from Nestlé's website at www.nestle.com/investors. You may find the information overwhelming for now, but try to spot the key principles that we have discussed in this chapter. It will get progressively easier as you gain familiarity with the elements of the financial statements.

Requirements

1. What method does Nestlé use to measure its inventories?
2. What would be the effect of Nestlé adopting FIFO to measure the value of its sundry supplies (previously measured under weighted average method) assuming rising inventory prices?
3. Three important pieces of inventory information are the cost of inventory on hand, (a) the cost of goods sold, and (b) the cost of inventory purchases. Assume that the actual cost of inventory is equal to the fair value of the inventory, at CHF 8,401 million and CHF 8,153 million at December 31, 2016 and 2015, respectively. Identify or compute items (a) and (b) for Nestlé for the year ended December 31, 2016.
4. Did Nestlé's gross profit margin percentage (use sales and cost of goods sold) on company sales improve or deteriorate in the year ended December 31, 2016, compared to the previous year?
5. Calculate Nestlé's inventory turnover for the years ended December 31, 2015, and December 31, 2016. Did inventory turnover rise or fall? How does this affect your analysis of Nestlé? What additional information would help you make a better judgment on Nestlé's inventory management practices? (Assume that the value of inventory at December 31, 2014, is CHF 9,170 million.)

Group Project

LO 4

■ writing assignment

(*Learning Objective 4: Comparing companies' inventory turnover ratios*) Obtain the annual reports of 10 companies, two from each of five different industries. Most companies' financial statements can be downloaded from their websites.

1. Compute each company's gross profit percentage and rate of inventory turnover for the most recent two years. If annual reports are unavailable or do not provide enough data for multiple-year computations, you can gather financial statement data from *Moody's Industrial Manual*.

2. How well does each of your companies compare to the other company in its industry?
What insight about your companies can you glean from these ratios?
3. Write a memo to summarize your findings, stating whether your group would invest in each of the companies it has analyzed.

Quick Check Answers

-
- | | | |
|-------------|-------------|-------------|
| 1. <i>c</i> | 2. <i>b</i> | 3. <i>d</i> |
|-------------|-------------|-------------|
4. *c* $[(15 \times \$6) + (25 \times \$7)]$
5. *b* $(15 \times \$6) + (10 \times \$7)$
6. *c* $25 \times [(\$90 + \$280 + \$90) \div 65]$
- | | | |
|-------------|-------------|-------------|
| 7. <i>c</i> | 8. <i>b</i> | 9. <i>d</i> |
|-------------|-------------|-------------|
10. *b* $(\$127,000 - \$75,000)$
11. *b* $(\$148,000 + \$28,000 - \$16,000)$
12. *c* $(\$28,000 + \$93,000 + \$3,000 - \$7,000 - \$35,000)$
13. *b* $(\$184,000 - \$82,000)/\$184,000$
14. *a* $[\$82,000 \div (\$28,000 + \$35,000)/2]$
15. *c* $\$120,000 + \$270,000 - [\$460,000 \times (1 - 0.40)]$
16. *b*

MyLab Accounting

For online homework, exercises, and problems that provide you with immediate feedback, please visit www.myaccountinglab.com.

APPENDIX 6A

ACCOUNTING FOR INVENTORY IN THE PERIODIC SYSTEM

In the periodic inventory system, the business keeps no running record of the merchandise. Instead, at the end of the period, the business counts inventory on hand and applies the unit costs to determine the cost of ending inventory. This inventory figure appears on the Balance Sheet and is used directly to calculate cost of goods sold.

Recording Transactions in the Periodic System

In the periodic system, throughout the period, the Inventory Account carries the beginning balance left over from the preceding period. During the period, the business records purchases of inventory in the *Purchases* account (an expense account). For inventory sales, only the revenue and considerations received (or receivable) are recorded.

At the end of the period, the purchases are added to the Inventory Account. A stocktake is then performed to determine the ending inventory balance. Any stock that is not present is *assumed* to be sold and charged to the cost of goods sold account. These end-of-period entries can be made during the closing process (see Chapter 3).

Exhibit 6A-1 illustrates the Inventory Accounting in a periodic system. Let's assume Jean-Luc Products (JLP) started the period with a beginning inventory balance of \$100,000. During the year, JLP made purchases on credit, totaling \$560,000. JLP's total sales for the year were \$900,000, all on credit. A stocktake at the end of the period revealed that JLP has an inventory ending balance of \$120,000.

Exhibit 6-A1 Recording and Reporting Inventories: Periodic System**PANEL A—Recording Transactions and the T-accounts**

A1				
	A	B	C	D
1	1.	Purchases	560,000	
2		Accounts Payable		560,000
3		<i>Purchased inventory on account.</i>		
4				
5	2.	Accounts Receivable	900,000	
6		Sales Revenue		900,000
7		<i>Sold inventory on account.</i>		
8				
9	3.	End-of-period entries to update Inventory and record Cost of Goods Sold:		
10	a.	Inventory	560,000	
11		Purchases		560,000
12		<i>Transferred purchases to inventory.</i>		
13				
14	b.	Cost of Goods Sold	540,000	
15		Inventory		540,000
16		<i>Set up ending inventory based on physical count.</i>		
17				

The T-accounts show the following:

Purchases		Inventory		Cost of Goods Sold	
Purchases 560,000	Inventory 560,000	Beg. bal. 100,000	COGS 540,000	Inventory 540,000	
End. Bal. 0		Purchases 560,000			
		End. Bal. 120,000			

PANEL B—Reporting in the Financial Statements**Income Statement (Partial)**

Sales revenue.....	\$900,000
Cost of goods sold:	
Beginning inventory.....	\$ 100,000
Purchases.....	560,000
Goods available.....	660,000
Ending inventory.....	(120,000)
Cost of goods sold.....	540,000
Gross profit.....	\$360,000

Ending Balance Sheet (Partial)

Current assets:	
Cash.....	\$ XXX
Short-term investments.....	XXX
Accounts receivable.....	XXX
Inventory.....	120,000
Prepaid expenses.....	XXX

APPENDIX ASSIGNMENTS

Short Exercises

S6A-1. (Recording inventory transactions in the periodic system) Paxton Technologies began the year with inventory of \$460. During the year, Paxton purchased inventory costing \$1,270 and sold goods for \$3,400, with all transactions on account. Paxton ended the year with inventory of \$630. Journalize all the necessary transactions under the periodic inventory system.

S6A-2. (Computing cost of goods sold and preparing the Income Statement—periodic system) Use the data in Short Exercise 6A-1 to do the following for Paxton Technologies:

► Requirements

1. Post to the Inventory and Cost of Goods Sold accounts.
2. Compute cost of goods sold by using the cost-of-goods-sold model.
3. Prepare the Income Statement of Paxton Technologies through gross profit.

Exercises MyLab Accounting

Select exercises can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select exercises using MyLab Accounting.

Group A

E6A-3A. (Computing amounts for various inventory methods—periodic system) Suppose Haley Corporation's inventory records for a particular computer chip indicate the following at July 31:

Jul 1	Beginning inventory	6 units @ \$60 = \$360
8	Purchase.....	3 units @ \$60 = 180
15	Purchase.....	14 units @ \$70 = 980
26	Purchase.....	2 units @ \$80 = 160

The physical count of inventory at July 31 indicates that seven units of inventory are on hand.

► Requirements

Compute ending inventory and cost of goods sold, using each of the following methods:

1. Specific unit cost, assuming two \$60 units and five \$70 units are on hand
2. Average cost (round average unit cost to the nearest cent)
3. First-in, first-out
4. Last-in, first-out

E6A-4A. (Journalizing inventory transactions in the periodic system; computing cost of goods sold) Use the data in Exercise 6A-3A.

► Requirements

Journalize the following for the periodic system:

1. Total July purchases in one summary entry. All purchases were on credit.
2. Total July sales in a summary entry. Assume that the selling price was \$298 per unit and that all sales were on credit.
3. July 31 entries for inventory. Haley uses LIFO. Post to the Cost of Goods Sold T-account to show how this amount is determined. Label each item in the account.
4. Show the computation of cost of goods sold by the cost-of-goods-sold model.

Group B

E6A-5B. (Computing amounts for various inventory methods—periodic system) Suppose Daxton Corporation's inventory records for a particular computer chip indicate the following at December 31:

Dec 1	Beginning inventory	7 units @ €62 = € 434
8	Purchase.....	5 units @ €62 = 310
15	Purchase.....	14 units @ €72 = 1,008
26	Purchase.....	3 units @ €82 = 246

The physical count of inventory at December 31 indicates that nine units of inventory are on hand.

► Requirements

Compute ending inventory and cost of goods sold, using each of the following methods:

1. Specific unit cost, assuming four €62 units and five €72 units are on hand
2. Average cost (round average unit cost to the nearest cent)
3. First-in, first-out
4. Last-in, first-out

E6A-6B. (*Journalizing inventory transactions in the periodic system; computing cost of goods sold*) Use the data in Exercise 6A-5B.

► Requirements

Journalize the following for the periodic system:

1. Total December purchases in one summary entry. All purchases were on credit.
2. Total December sales in a summary entry. Assume that the selling price was €318 per unit and that all sales were on credit.
3. December 31 entries for inventory. Daxton uses LIFO. Post to the Cost of Goods Sold T-account to show how this amount is determined. Label each item in the account.
4. Show the computation of cost of goods sold by the cost-of-goods-sold model.

Problems MyLab Accounting

Select problems can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select problems using MyLab Accounting.

Group A

P6A-7A. (*Computing cost of goods sold and gross profit on sales—periodic system*) Assume a Watercrest outlet store began July 20X6 with 50 units of inventory that cost \$16 each. The sale price of these units was \$70. During July, the store completed the following inventory transactions:

			Units	Unit Cost	Unit Sale Price
Jul	3	Sale	18	\$18	\$70
	8	Purchase	80	19	71
	11	Sale	32	18	70
	19	Sale	4	19	72
	24	Sale	36	19	72
	30	Purchase	21	20	72
	31	Sale	5	19	72

► Requirements

1. Determine the store's cost of goods sold for July under the periodic inventory system. Assume the FIFO method.
2. Compute gross profit for July.

P6A-8A. (*Recording transactions in the periodic system; reporting inventory items in the financial statements*) Accounting records for Decadent Desserts, Inc., yield the following data for the year ended December 31, 20X6 (amounts in thousands):

Inventory, December 31, 20X5	\$ 580
Purchases of inventory (on account)	2,240
Sales of inventory—70% on account; 30% for cash	3,600
Inventory at the lower of FIFO cost or NRV, December 31, 20X6	700

► Requirements

1. Journalize Decadent Desserts' inventory transactions for the year under the periodic system. Show all amounts in thousands.
2. Report ending inventory, sales, cost of goods sold, and gross profit on the appropriate financial statement (amounts in thousands). Show the computation of cost of goods sold.

Group B

P6A-9B. (Computing cost of goods sold and gross profit on sales—periodic system) Assume a Championship outlet store began January 20X6 with 52 units of inventory that cost €20 each. The sale price of these units was €75. During January the store completed these inventory transactions:

		Units	Unit Cost	Unit Sale Price
Jan 3	Sale	19	€20	€75
8	Purchase.....	75	21	73
11	Sale	33	20	75
19	Sale	3	21	77
24	Sale	40	21	77
30	Purchase.....	20	22	74
31	Sale	7	21	77

► Requirements

1. Determine the store's cost of goods sold for January under the periodic inventory system. Assume the FIFO method.
2. Compute gross profit for January.

P6A-10B. (Recording transactions in the periodic system; reporting inventory items in the financial statements) Accounting records for Sinful Desserts, Inc., yield the following data for the year ended December 31, 20X6 (amounts in thousands):

Inventory, December 31, 20X5	€ 520
Purchases of inventory (on account).....	2,100
Sales of inventory—75% on account, 25% for cash.....	4,000
Inventory at the lower of FIFO cost or market, December 31, 20X6.....	680

► Requirements

1. Journalize Sinful Desserts' inventory transactions for the year under the periodic system. Show all amounts in thousands.
2. Report ending inventory, sales, cost of goods sold, and gross profit on the appropriate financial statement (amounts in thousands). Show the computation of cost of goods sold.

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